My health @ EU
eHealth Digital Service Infrastructure
A service provided by the European Union

My Health @ EU
eHealth Digital Service Infrastructure
Electronic cross-border health services in the EU

NCPeH Components Specifications

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ABSTRACT
This document covers the technical specifications of the MyHealth@EU common components that have to be put in place by MyHealth@EU countries in order to allow for a secure and privacy-aware exchange of identifiable medical data. The document is presented as a consolidated specification of the already implemented MyHealth@EU solution. It furthermore serves as the foundation of the MyHealth@EU services technical specification, which is aimed at reusing as much of the work already done and to preserve much of the backwards compatibility between both initiatives.

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1 Introduction

This document covers the technical specifications of the MyHealth@EU common components that have to be put in place by MyHealth@EU countries in order to allow for a secure and privacy-aware exchange of identifiable medical data.
The document is presented as a consolidated specification of the already implemented MyHealth@EU solution. It furthermore serves as the foundation of the MyHealth@EU services technical specification, which is aimed at reusing as much of the work already done and to preserve much of the backwards compatibility between both initiatives.

1.1 eHealth Digital Service Infrastructure

The core principle of MyHealth@EU is to bridge existing national eHealth infrastructures instead of setting up a new, centralised European healthcare service network from scratch. To make this approach work technical, semantic and legal interoperability among European eHealth infrastructures must be achieved. This includes identity matters as well as security matters and information management issues within heterogeneous, distributed environments. None of the problems faced by MyHealth@EU is new or extremely challenging: cross-border data exchange is common practice in many domains, nearly every European country has use cases and processes for cross-enterprise health data exchange defined and decoupled security services spanning federated domains are well covered by international standards.
Therefore the challenge for the development of technical specifications for the MyHealth@EU building blocks is not to find a solution that works for the defined use cases. The challenge is to find a solution that

1) can evolve over the next years and lay ground for a seamless, secure and privacy-aware exchange of any kind of medical data across Europe,

2) can easily connect to the existing infrastructures without imposing unreasonable new risks on the privacy and integrity of existing data managing systems,

3) is flexible enough to be used in conjunction with different means of identification, authentication and authorisation to allow for any citizen and country to participate based on existing legal regulations and technical actualities and

4) is widely accepted and has the potential for reuse of software components and test tools.

1.2 MyHealth@EU Common Components Specification

The *MyHealth@EU Common Components Specification* is the technical baseline of MyHealth@EU: There MUST NOT be any normative statement on the shape and behaviour of any MyHealth@EU building block that is on a technically lower level than this specification. While specifications on identity management [Identity management Specifications] and security services [Security Services Specification] define the core concepts and guidelines for these issues, the *MyHealth@EU Common Component Specifications* defines the technical means that serve these concepts.

The scope of the *MyHealth@EU Common Components Specification* is determined by:

- **technical interoperability**: the specification does only cover aspects of cross-border data exchange that are indispensable for technical interoperability

- **shared responsibility**: whenever possible, only the shape of data exchanged "on the wire" is specified while the processes and systems for issuing and consuming these data objects are considered to be national responsibilities

The objective is to define the MyHealth@EU interoperability building blocks in a way that they can be implemented independently from each other by different vendors. The *MyHealth@EU Common Components Specification* builds upon the implementation independent technology view of [System Architecture Specifications]. It provides the normative specification of the MyHealth@EU services' operations by mapping them onto established standards.

1.3 Conventions

The keywords MUST, SHOULD, MAY, SHOULD NOT and MUST NOT are used as defined in [RFC 2119].

For indicating the optionality of certain functionalities or data fields the following keywords are used:

- "**R**" or "Required": Functionalities and data fields marked as "R" MUST be provided.
- The notation "**R+**" is used to indicate that MyHealth@EU is stricter on the mandatory nature of the functionality or data field than the underlying base standard.
- "**O**" or "Optional": Functionalities and data fields marked as "O" MAY be omitted.
- "**R/O**": Data fields marked as "R/O" are conditionally mandatory.
- "**X**": Functionalities and data fields marked as "X" MUST NOT be used.

1.4 Organisation of this Document

This document consists of four main parts in order to separate different levels of abstraction and to allow different groups of readers to easily access the information that is relevant for them without having to read the whole document:

- Chapter 2 of this document specifies the functionality of the **MyHealth@EU Services** by outlining the data elements that have to be exchanged in order to implement the defined service operations. This MyHealth@EU functional service specification is targeted at system architects who are responsible for connecting HCPOs and existing national infrastructures to MyHealth@EU components.
- Chapter 3 specifies the implementation of the MyHealth@EU services by providing the **MyHealth@EU Mappings** of the service operations. These specifications describe how the MyHealth@EU services' operations are mapped onto existing standards and healthcare profiles. This part is targeted at software designers and developers who are responsible for the implementation of MyHealth@EU services and for their integration with existing national infrastructure components.
- Chapter 4 defines the **MyHealth@EU communication and messaging infrastructure**. It describes the protocols to be used for establishing the MyHealth@EU network of trusted nodes and specifies the common formats for messages. Chapter 4 is mainly targeted at systems administrators who are responsible for the configuration of web service platforms and network nodes.
- Chapter 5 specifies the syntax and semantics of the **assertions and certificates** that are used by the MyHealth@EU services. This chapter is

targeted at security administrators who are responsible for the management of certificates and the configuration of security services.

Sample messages and the WSDLs for all MyHealth@EU services' interfaces are provided in an appendix to this document.

2 MyHealth@EU Services Functional Specification

The MyHealth@EU architecture is based on a service-oriented paradigm. [NCPeH Architecture Specification] defines the MyHealth@EU services and service interfaces and shows how these interact with each other in order to implement the MyHealth@EU use cases on patient summary, ePrescription and original clinical documents (OrCD).

This chapter builds upon the technology viewpoint on the MyHealth@EU architecture by further refining the functional specification of the MyHealth@EU services' interfaces. For each service operation the input and output as well as the requested effect and the preconditions for a success scenario are defined.

2.1 MyHealth@EU Service-Oriented Architecture

The design of the general architecture of MyHealth@EU and the design of the MyHealth@EU services is based on the following basic assumptions:

1. The design uses the service-oriented paradigm.
2. All services are passive, the service consumers and service providers communicate synchronously.
3. All MyHealth@EU medical data as well as all patient and HP identity data is administered in autonomous systems. Any exchange of these data is mediated by national gateways following a B2B paradigm.
4. There are no central services except the ones needed for the operation of a secure MyHealth@EU infrastructure; the federation of national gateways is implemented via a »circle of trust«.

2.1.1 Service Roles

The service-oriented paradigm distinguishes among three roles: service provider, services consumer, and service registry. The service provider offers a service that is used by the service consumer. The service provider can publish a description of the service in a service registry.

MyHealth@EU services are implemented as Web Services whose interfaces are specified with the Web Service Description Language [W3C WSDL 1.1]. MyHealth@EU Web services are passive. Communication between service consumer and service provider is always initiated by the service consumer. The service provider is passive and reacts to requests from the service consumer.

The communication between service consumer and services provider is synchronous. The service consumer's control flow is suspended until the service provider has processed the service consumer's request. Service consumer and provider communicate over the Internet using XML-based SOAP messages transported via the HTTP protocol (see chapter 4.3 for details).

Each MyHealth@EU service is operated under the responsibility of a National Contact Point (NCP). The role of the service consumer is always taken by the NCP of the country of care (country B). The role of the service provider is always taken by the NCP of the country of the patient's affiliation (country A).

A service registry is not used. Instead it is assumed that service descriptions and service location information is made available for service consumers by organisational means and static local directory services (see section 2.1.4). As with other centrally managed data (e. g. value set catalogues and trusted certificates), the mechanisms for securely distributing

this information among NCPs are out of the scope of this document and will instead be covered by the MyHealth@EU documents on security policies and pilot operations.

2.1.2 Trusted federation of national Contact Points

National Contact Points (NCPs) are implemented federally in order to allow for a virtual integration of autonomous sources of medical data and identity information [OFW-NCPeH]. The independence of the information sources is preserved as all access to an information source is mediated through the MyHealth@EU services which are implemented at NCP level. This complies to a Business-to-Business paradigm where end users and data sources are decoupled by enterprise level entry services that map external requests onto internal operations and vice versa. Part of this mediation is the brokerage of trust that is achieved by matching security objects of the NCP-to-NCP trust domain into a local trust domain and vice versa.

The administration of the identities of patients and healthcare professionals (HPs) is decentralized. Authentication of a HP can only be done within the country of care that recognizes this HP. The existence of a treatment relationship between a patient and a HP can only be attested within the legal framework of the point of care (PoC). With MyHealth@EU HP authentication and treatment relationship attestment are therefore performed within the country of care (e. g. by using an existing Identity Provider service of the national infrastructure). A brokerage of the HP authentication and treatment relationship confirmation into the MyHealth@EU domain is performed in a way that the NCP at country B confirms the respective claims and maps them onto a unified syntax and semantics that can be processed by the NCP of country A.

The MyHealth@EU "Circle of Trust" consists of pairs of mutually trusted consuming and providing gateways. A *consuming gateway* provides the computing environment for the operation of an MyHealth@EU service consumer and acts as the exit point from the national domain into the MyHealth@EU domain. A *providing gateway* operates the Web Service Endpoint (WSE) of the service provider and acts as the entry point from the MyHealth@EU domain into the national domain. Each gateway is operated under the legal responsibility of an NCP.

A basic assumption of MyHealth@EU is that mutual trust exists between NCPs. The authenticity and integrity of the services in the gateway-mediated NCP-2-NCP communication is secured with digital certificates. Examinations of the certificates and their exchange (with a limited number of NCPs) are to be synchronized among the MyHealth@EU service providers [Security Services Specification, NCPeH Architecture Specification].

2.1.3 MyHealth@EU Services Overview

To foster interoperability among European healthcare infrastructures without forcing countries to modify their running eHealth services, only NCP gateway-to-gateway interfaces are considered as "normative" with MyHealth@EU. These interfaces are provided by web services that use open, XML-based standards and transport protocols to exchange data between gateways.

The full list of MyHealth@EU web services acc. to [NCPeH Architecture Specification] is shown in Table 1.

Service	Mediated Data	Functional Specification
Identification Service	Patient identifiers and demographics	[Identity Management Specification]
Patient Service	Patient summary documents	[PS Functional requirements]

Order Service	ePrescription documents	[eP Functional requirements]
Dispensation Service	eDispensation documents	[eP Functional requirements]
OrCD Service	OrCD documents	[OrCD Functional requirements]

Table 1: MyHealth@EU Services Overview

2.1.4 Services Localisation

In MyHealth@EU, service discovery and location will be based on dynamic look-up [NCPeH Architecture Specification]. Each NCP MUST describe its service addresses and certificates in a centrally managed location table that complies to the *MyHealth@EU Service Metadata Publisher* format as specified in [Service Location and Capability Lookup Profile] (see section 2) of this document.

Each NCP holds a copy of the other NCP's location tables as part of its internal configuration.

2.1.5 General Considerations for Service Operations

In order to successfully operate the MyHealth@EU use cases on patient summary and ePrescription, the following preconditions MUST be met:

1) The service consumer MUST be able to locate the service provider. The respective end point addresses and certificates MUST be provided through a [Service Location and Capability Lookup Profile] (see section 2). Up-to-date copies of all service providing NCP's Service Metadata Publisher files MUST be available to the service consumer.
2) A secure channel MUST have been established between service consumer and service provider nodes (see section 4.1 for the establishment of the MyHealth@EU Trusted Node Infrastructure).
3) The service provider MUST be able to verify the authenticity of the service consumer and vice versa. This requires that service consumer and service provider only make use of digital certificates that can be verified by the counterpart (see [SAML Profile] for the definition of the respective certificate profiles).
4) The requesting HP MUST have been authenticated in the country of care (see [Identity Management Specification] for details on HP authentication). The service provider MUST be able to verify the attesting HP identity assertion (see [SAML Profile] for the specification of the HP Identity Assertion and [X.509 Certificate Profiles]. for the certificate profile of the NCP signature certificate that MUST be used for attesting the successful authentication of an HP).

Further general requirements for secure and privacy-aware operations that MUST be considered are defined in [System Architecture Specification] and [Security Services Specification].

Failures during a service's operation or faulty conditions for using a service MUST be handled. Part of this fault handling is that a respective audit trail entry MUST be written at all NCPs who are aware of the fault. See section 4.4 for details.

3 MyHealth@EU Service Implementation

3.1 IHE-based Services

The MyHealth@EU service interface defines the semantics of identification and data sharing operations for European health services. "On the wire" the service operations are

implemented by SOAP messages that are exchanged between an initiating gateway (operated by NCP-B) and a providing gateway (operated by NCP-A).
This chapter specifies the mapping of MyHealth@EU service operations onto standardised messages and as such is the normative implementation guideline for the MyHealth@EU-facing NCP interface.

3.1.1 Conventions and Restrictions

3.1.1.1 NCP: A Standards based Implementation

The MyHealth@EU NCP2NCP interface is based on the IHE X* family of Interoperability Profiles and additionally utilises a set of supporting profiles (Figure 6).

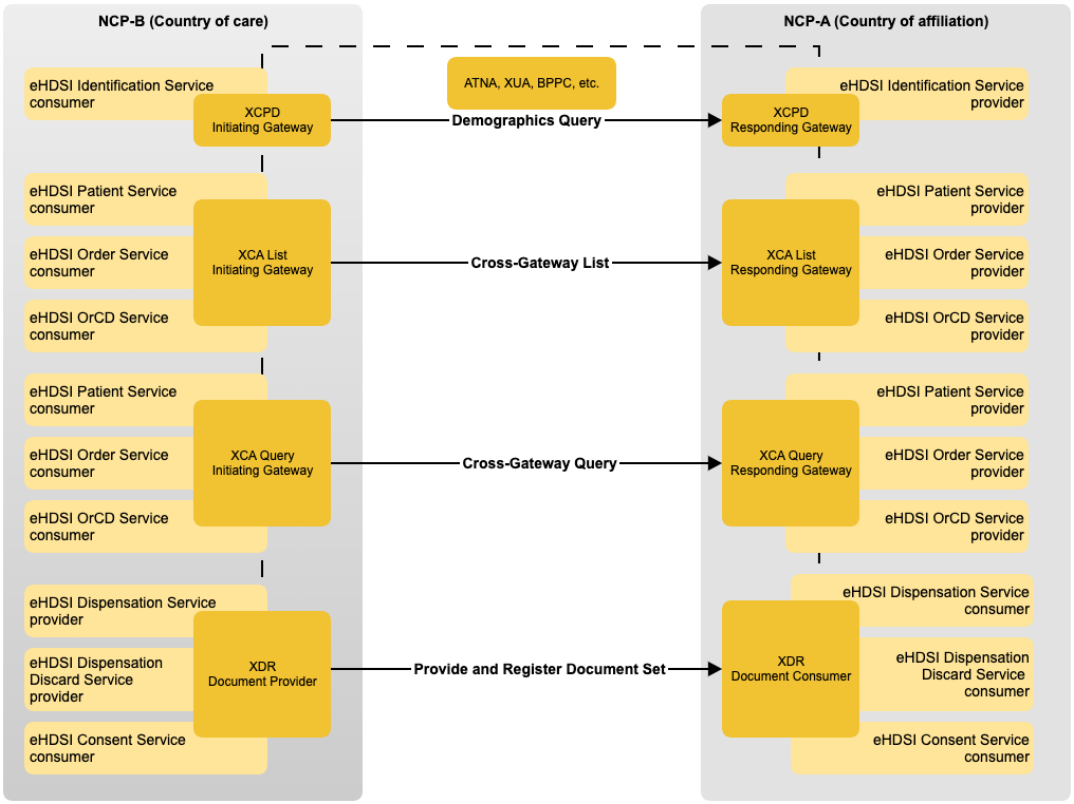


Figure 6: IHE actors and transactions that are profiled by MyHealth@EU

The IHE profiles that are implemented for MyHealth@EU NCP-to-NCP data exchange are:

- Cross-Community Access (XCA) incorporating a modified getAll() transaction
- Cross-Enterprise Document Reliable Interchange (XDR)
- Cross-Community Patient Discovery (XCPD)

The MyHealth@EU service specifications that build on these IHE profiles introduce various extensions and restrictions on IHE actor and transaction definitions in order to properly cover the MyHealth@EU use cases and to align with the MyHealth@EU security framework:

- Registry query and repository retrieve transactions are conflated to a single list() operation (see section 3.1.2).

- Additional error messages are defined that cover specific failure conditions of the MyHealth@EU use cases on Patient Summary and ePrescription.
- Warning messages are introduced (see section 3.1.4) to allow for notifications on specific environmental conditions that apply to a country and that MAY affect the interpretation of data in another country.
- The optionality of data fields is aligned to European privacy regulations.
- The application of security measures and the contents of the SOAP security header are specified normatively (see section 4.3.5).

3.1.1.2 IHE Cross Community Access (XCA)

IHE XCA defines message interchange formats for listing and retrieving patient related documents. It is based on a stored query model[1] where the requestor provides attribute-value pairs in the query. It is up to the responding site to map the query attributes onto its internal registry information model. For documents matching the provided attributes' values, the document identifiers (and minimum metadata) are provided. The initiator of the original request may then chose a subset of these listed documents to issue a simple retrieve based on the document identifiers.

At this point IHE XCA only supports scenarios where a query to request a list is first performed by returning document metadata (»LeafClass«) or just document IDs (»ObjectRef«), followed by an additional document retrieve operation.

In contrast MyHealth@EU is based on a pattern, that allows for easier handling of dynamic data and keeps the NCP-A stateless [PS Functional requirements]. For MyHealth@EU the retrieval of a single patient summary or a patient's set of ePrescriptions is performed by a single fetch() operation which returns all objects that match a given filter query within the response message. Error Handling

Failures during operation execution can be of different kinds; e.g. they may be caused by syntactic mismatches, insufficient access rights, country-A component failures, or protocol failures. MyHealth@EU makes use of three different error reporting mechanisms in order to allow for a better handling of errors on the appropriate level of abstraction (see Figure 7):

- SOAP faults: The standard SOAP fault mechanism is used for failures that originate in the encoding of the SOAP message or the contents of the SOAP header. It is assumed that the respective errors are discovered during the processing of the message at the MyHealth@EU communication tier of the NCP-A and that they mainly address failures that originate at NCP-B. Typical examples of such errors are missing security token or usage of undefined attributes within security token.
- Error Messages in the SOAP response body: Error reporting mechanisms of the business level protocol (e. g. XCA) are used for failures that are discovered during the business-level processing of security token and SOAP body elements. These errors may as well be discovered during policy enforcement at the NCP as during the processing of the request within the national infrastructure. The failure usually either originates at the Point of Care in country B or at the national infrastructure in country A. These errors SHOULD be reported to the HP in country B as it is assumed that either the HP or the patient MAY be able to take action to successfully re-issue the request. Typical examples of such errors are missing consents and temporary component failures in country A.
- Error messages related to the creation of the document content: There may be cases where failure to access certain systems within a national infrastructure may result in some elements of clinical information missing (e.g. in a patient summary).

[1] IHE XCA – as IHE XDS.b – does not exchange full query statements but only query arguments that are filled into predefined slots of a database stored query at the service side.

These clinical content errors should be conveyed within the document content. The SOAP body transactions and SOAP header were exchanged without errors at the lower two levels.

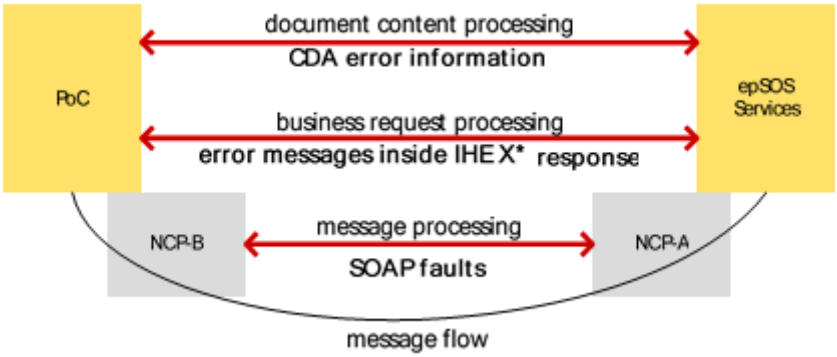


Figure 7: MyHealth@EU error handling

A list of all SOAP fault codes and conventions to be followed for transmitting SOAP faults are provided in section 4.4 of this document. Business level error messages and their proposed processing are provided in the "response message" sections of the MyHealth@EU service specifications. Lower level communication and protocol failures are covered in section 4.4.1 of this document.

3.1.1.3 Information Messages and Warnings

MyHealth@EU implements medical data sharing among different legal and technical environments. This might lead to scenarios where the NCP at the patient's country of affiliation MAY wish to send further information on the data collection procedure or on the source environment together with the data to the HP in the country of care. An example for this is a notice on automatically collected data that was not approved by an HP. Even an uncertain state of a request's fulfilment – e.g. NCP was not able to access all relevant data sources – MUST be reported to the data consumer in order to provide a correct semantic context for the provided data.

To allow for this exchange of context information, all ebXML based MyHealth@EU messages provide the ability to include an <rs:RegistryErrorList/> element (with an success indicator) for transmitting information and warnings together with provided medical data. Warnings that only affect the contents of a single document are reported in an explicit clinical statement within that document.

3.1.1.4 Object Identifier

In the absence of an official MyHealth@EU OID [ISO OID] a temporary OID is assigned as the root OID 1.3.6.1.4.1.12559.11.10.1.3 for MyHealth@EU.

- (Branch 1 was allocated to MyHealth@EU CDA templates: 1.3.6.1.4.1.12559.11.10.1.3.**1**).
- Branch 2 (1.3.6.1.4.1.12559.11.10.1.3.**2**) was allocated to Common Components

The following branches are defined for codes and code systems:

OID Branch	Description
1.3.6.1.4.1.12559.11.10.1.3.2.1	Patient Identification related codes and code systems
1.3.6.1.4.1.12559.11.10.1.3.2.2	HP/HCPO identification, authentication, authorisation related codes and code systems
1.3.6.1.4.1.12559.11.10.1.3.2.3	Medical data sharing related codes and code systems

1.3.6.1.4.1.12559.11.10.1.3.2.4	Consent encoding and management related codes and code systems

The following branches are defined for MyHealth@EU CDA templates:

OID Branch	Description
1.3.6.1.4.1.12559.11.10.1.3.1.1.1	ePrescription
1.3.6.1.4.1.12559.11.10.1.3.1.1.2	eDispensation
1.3.6.1.4.1.12559.11.10.1.3.1.1.3	Patient Summary
1.3.6.1.4.1.12559.11.10.1.3.1.1.8	eHDSI OrCD Laboratory Result
1.3.6.1.4.1.12559.11.10.1.3.1.1.9	eHDSI OrCD Hospital Discharge Report
1.3.6.1.4.1.12559.11.10.1.3.1.1.10	eHDSI OrCD Medical Imaging Report
1.3.6.1.4.1.12559.11.10.1.3.1.1.11	eHDSI OrCD Medical Images

For a full list of all defined OIDs see appendix 6.2.2 of this document.

3.1.1.5 Namespaces

XML namespace prefixes are used in this document to stand for their respective namespaces as follows.

Prefix	Namespace
ehealth	urn:ehealth:v1
soapenv	http://www.w3.org/2003/05/soap-envelope
wsse	http://docs.oasis-open.org/wss/2004/01/oasis-200401-wss-wssecurity-secext-1.0.xsd
saml	urn:oasis:names:tc:SAML:2.0:assertion
xacml	urn:oasis:names:tc:xacml:2.0:policy:schema:os
hl7v2	urn:hl7-org:v2
hl7v3	urn:hl7-org:v3
xds	urn:ihe:iti:xds-b:2007
xs	http://www.w3.org/2001/XMLSchema
xsi	http://www.w3.org/2001/XMLSchema-instance
rimext	urn:ihe:iti:xds-ebrim:extensions:2010[2]
query	urn:oasis:names:tc:ebxml-regrep:xsd:query:3.0
rim	urn:oasis:names:tc:ebxml-regrep:xsd:rim:3.0
rs	urn:oasis:names:tc:ebxml-regrep:xsd:rs:3.0
lcm	urn:oasis:names:tc:ebxml-regrep:xsd:lcm:3.0
tsl	http://uri.etsi.org/02231/v2#

3.1.2 MyHealth@EU Identification Service

The *MyHealth@EU Identification Service* is used to discover a valid patient identifier from an ID assigning authority by providing given identifiers and/or demographic data that is sufficient for patient identification.



Figure 8 - Patient Identification Service Interface

[2] IHE will be asked to block this namespace for MyHealth@EU extensions in order to avoid conflicts with forthcoming IHE extensions to the ebRIM.

The implementation of the MyHealth@EU Identification Service is based on the standard

- HL7 IS: HL7 V3 Identification Service

and is an extension to the IHE profile

- XCPD: IHE Cross-Community Patient Discovery [IHE XCPD]

3.1.2.1 findIdentityByTraits() Operation

The *IHE XCPD Cross-Gateway Patient Discovery* transaction – as for the semantics and syntax of its content - is based on *HL7 Patient Registry Find Candidates Query* (PRPA_IN201305UV02) interaction type and used also for the [IHE PIX/PDQ v3] transactions.

3.1.2.1.1 Event Identification

The *findIdentityByTraits()* request is initiated by an HP in the country of care for the identification of a foreign patient. The respective request message conforms to the *Patient Registry Find Candidates Query* (PRPA_IN201305UV02) interaction type as profiled by the IHE XCPD *Cross-Gateway Patient Discovery* transaction [IHE XCPD].
For the HL7 transmission wrapper and the *HL7 Control Act* the conventions identified in the IHE PIX/PDQV3 supplement appendix O [IHE PIX/PDQ v3] and the changes from the XCPD supplement appendix O MUST be followed.

In addition the following MyHealth@EU-specific restrictions apply:

- <receiver/> MUST refer to NCP-A. Other sub-elements than the device-identifier that holds the OID of NCP-A MUST be ignored by the service provider and SHOULD NOT be provided by the service consumer. <sender/> MUST refer to NCP-B. Other sub-elements than the device-identifier that holds the OID of NCP-B MUST be ignored by the service provider and SHOULD NOT be provided by the service consumer.
- Asynchronous operations MUST NOT be used. According to [PS Functional requirements] all message interchange in MyHealth@EU MUST be synchronous.
- *Demographic Query Only mode* or *Shared/national Patient Identifier Query and Feed mode*
- MUST be used. Other modes defined in [IHE XCPD] MUST NOT be used.
- The *health data locator option* as defined in section 27.2.1 of [IHE XCPD] MUST NOT be used. Where indication of support for the *health data locator option* is required in responses, the service provider MUST provide the value "NotHealthDataLocator".
- The *revoke option* as defined in section 27.2.2 of [IHE XCPD] MUST NOT be used.
- Correlations MUST NOT be cached by the service provider. The respective syntax elements described in section 3.55.4.1.2 of [IHE XCPD] MUST NOT be used.
- Reverse Cross-Gateway Queries MUST NOT be used. The *homeCommunityId* and *community patient id assigning authority* arguments SHOULD be set to the OID of the responding NCP (NCP-A) in query requests

3.1.2.1.2 Restrictions on the Use of Traits

For a *findIdentityByTraits()* request only the following traits or a subset of these MUST be used. Service providers SHOULD reject requests that contain other traits than the ones listed below.

Identity Trait	Source	Usage Convention (if provided)

LivingSubjectID	Personal ID Card	SHOULD contain zero or more living subject Id. When present, it shall contain both an assigning authority identifier (root) and individual ID (extension). If multiple subject IDs are given for the same patient, each identifier MUST be provided as a dedicated <LivingSubjectID/> element and some of them might be optional.
LivingSubjectName	Personal ID Card	Family name and given name MUST both be given if no Liv- ingSubjectID is provided. Otherwise this query parameter is optional.
LivingSubjectBirthTime	Personal ID Card	Birth date MUST be given if no LivingSubjectID is provided. Otherwise this query parameter is optional. If given this parameter MUST be encoded as "YYYY[MM[DD[HHMM[SS[.S[S[S[S]]]]]]]][+/-ZZZZ]" with year, month and day being mandatory.
LivingSubjectGender		MUST be "M", "F" or "UN"
PatientAddress	Personal ID Card	SHOULD contain country and city.

For detailed information on the encoding of these traits and their optionality for the XCPD modes see [IHE XCPD].

The set of required traits for patient identification are defined by each eHNCP-A according the [Service Location and Capability Lookup Profile]; Service Metadata Publisher EHEALTH-107.

3.1.2.1.3 Use of Pseudonyms and Temporal Identifiers

A country MAY wish to protect its patients' privacy by negotiating an MyHealth@EU shared identifier from a pseudonymous or temporal national patient identifier. In this case *Shared/national Patient Identifier Query and Feed mode* MUST be used. On successful identification within country A the response message MUST at least provide the patient's date of birth in order to allow the HP in country B to verify the accuracy of the identification.

3.1.2.1.4 Patient Authentication

The *MyHealth@EU Identification Service* allows for the identification of a patient. If a country requires an additional authentication of its citizens when they ask for medical care in another country, this country MUST define its own authentication service. The *MyHealth@EU Identification Service* findIdentityByTraits operation only provides the mechanism for piggybacking the exchange of transparent authentication data between NCPs. This is done by using two HL7v3 *instance identifier* within a single <LivingSubjectID/> element; one identifier is used for identifying the patient while the other one is used for authentication of the patient.
The service provider MUST be able to distinguish identifier and authentication object by their roots (assigning authorities).

3.1.2.1.5 Requested Accuracy of Matches

The *HL7 Patient Registry Find Candidates Query* allows to further refine the match criteria by setting the match algorithm and specifying a requested minimum degree of match for the provided traits.
As the respective <MatchCriterionList> element is optional with the HL7 schema, it SHOULD NOT be used for the MyHealth@EU wave 1. If present, the minimum requested

match degree SHOULD be set to an integer value of "100". In both cases the responding service SHOULD only respond with identity data of patients who fully match all provided traits. Returning multiple candidates' identity trails SHOULD be avoided for privacy reasons.

3.1.2.1.6 Example Request Message

The following excerpt from a *findIdentityByTraits* request message shows the IHE XCPD profile of the HL7 PRPA_IN201305UV02 interaction type. The request message can be used to retrieve the identifier of a patient who identified himself with his electronic health card and date of birth.

```
<soapenv:Envelope>
<soapenv:Header> … >/soapenv:Header>
<soapenv:Body>
<hl7v3:PRPA_IN201305UV02 xmlns:xsi="...">
   <hl7v3:id root="36E66A20-1DD2-11B2-90FA-80CE4046A4A7"/>
   <hl7v3:creationTime value="20100304120000"/>
   <hl7v3:interactionId root="2.16.840.1.113883.1.6" extension="PRPA_IN201305UV02"/>
   <hl7v3:processingCode code="P"/>
   <hl7v3:processingModeCode code="T"/>
   <hl7v3:acceptAckCode code="NE"/>
   <hl7v3:receiver typeCode="RCV">
      <hl7v3:device classCode="DEV" determinerCode="INSTANCE">
         <hl7v3:id root="1.2.840.114350.1.13.999.234"/>
      </hl7v3:device>
   </hl7v3:receiver>
   <hl7v3:sender typeCode="SND">
      <hl7v3:device classCode="DEV" determinerCode="INSTANCE">
         <hl7v3:id root="1.2.840.114350.1.13.999.567"/>
      </hl7v3:device>
   </hl7v3:sender>
   <hl7v3:controlActProcess  classCode="CACT" moodCode="EVN">
      <hl7v3:code code="PRPA_TE201305UV02" codeSystem="2.16.840.1.113883.1.6"/>
      <hl7v3:queryByParameter>
         <hl7v3:queryId root="1.2.840.114350.1.13.28.1.18.5.999" extension="18204"/>
         <hl7v3:statusCode code="new"/>
         <hl7v3:responseModalityCode code="R" />
         <hl7v3:responsePriorityCode code="I"/>
         <hl7v3:parameterList>
            <hl7v3:livingSubjectBirthTime>
               <hl7v3:value value="19600422"/>
               <hl7v3:semanticsText/>
            </hl7v3:livingSubjectBirthTime>
            <hl7v3:livingSubjectId>
               <!-- German electronic healthcard card number (Serial Number) -->
               <hl7v3:value root="1.2.276.0.76.4.8" extension="1234567890"/>
               <hl7v3:semanticsText/>
            </hl7v3:livingSubjectId>
         </hl7v3:parameterList>
      </hl7v3:queryByParameter>
   </hl7v3:controlActProcess>
</hl7v3:PRPA_IN201305UV02>
</soap:body>
</soap:envelope>
```

The following example shows the mapping of Czech EHIC data elements onto a PRPA_TE201305UV02 control act.




```
<hl7v3:controlActProcess   classCode="CACT" moodCode="EVN">
 <hl7v3:code code="PRPA_TE201305UV02" codeSystem="2.16.840.1.113883.1.6"/>
 <hl7v3:queryByParameter>
  <hl7v3:queryId root="1.2.840.114350.1.13.28.1.18.5.999" extension="18204"/>
  <hl7v3:statusCode code="new"/>
  <hl7v3:responseModalityCode code="R" />
  <hl7v3:responsePriorityCode code="I"/>
  <hl7v3:parameterList>
   <hl7v3:livingSubjectBirthTime>
    <hl7v3:value value="19501201"/>
    <hl7v3:semanticsText/>
   </hl7v3:livingSubjectBirthTime>
   <hl7v3:livingSubjectId>
    <!-- European Health Insurance Card Serial Number -->
    <hl7v3:value root="......." extension="80203111990000000001"/>
                        <hl7v3:semanticsText/>
                    </hl7v3:livingSubjectId>
                    <hl7v3:livingSubjectName>
                        <hl7v3:value>
                            <hl7v3:family>NOVAK</hl7v3:family>
                            <hl7v3:given>JAN</hl7v3:given>
                        </hl7v3:value>
                        <hl7v3:semanticsText/>
                    </hl7v3:livingSubjectName>
                    <hl7v3:patientAddress>
                        <hl7v3:value>
                            <hl7v3:country>CZ</hl7v3:country>
                        </hl7v3:value>
                        <hl7v3:semanticsText/>
                    </hl7v3:patientAddress>
                </hl7v3:parameterList>
            </hl7v3:queryByParameter>
        </hl7v3:controlActProcess>
```

3.1.2.1.7 Active Participant Role ID Codes

The *MyHealth@EU Identification Service* provider shall respond with the *findIdentityByTraits* response message containing the patient identifier that is to be used for querying the identified patient's medical data. The *MyHealth@EU Identification Service* provider MUST verify that the requesting service user has sufficient rights to query for the identifier of the given patient. It is subject to the national security policy of the patient's country of affiliation, how multiple matches and matches with less than 100% accuracy are handled[3].

[3] The national security policy of the patient's country of affiliation always overrides service consumer minimum confidence level: for instance, if country A only accept 100% match but country B is requesting with minimum

In case of an error that relates to the transmission of the request or the processing of the MyHealth@EU security token, the *MyHealth@EU Identification Service* provider MUST respond with a fault message according to section 4.4 of this document.

3.1.2.1.8 Response Message (Full Success Scenario)

The *MyHealth@EU findIdentityByTraits response* content is based on *HL7 Patient Registry Find Candidates Query Response (PRPA_IN201306UV02)* interaction, as profiled by the *IHE XCPD Cross-Gateway Patient Discovery* result message.
For the HL7 transmission wrapper and the *HL7 Control Act* the conventions identified in the IHE PIX/PDQV3 supplement appendix O and the changes from the XCPD supplement appendix O MUST be followed.
In addition the following MyHealth@EU-specific restrictions apply:

- <receiver/> MUST refer to NCP-B. Other sub-elements than the device-identifier that holds the OID of NCP-B MUST be ignored by the service consumer and SHOULD NOT be provided by the service provider. <sender/> MUST refer to NCP-A. Other sub-elements than the device-identifier that holds the OID of NCP-A MUST be ignored by the service consumer and SHOULD NOT be provided by the service provider.
- Asynchronous operations MUST NOT be used. According to [PS Functional requirements] all message interchange in MyHealth@EU MUST be synchronous.
- Correlations MUST NOT be cached by the service provider. The respective syntax elements described in section 3.55.4.1.2 of [IHE XCPD] MUST NOT be used.
- The <processingCode/> MUST be set to "D" (debugging) for MyHealth@EU PPT environment. It MUST be set to "P" (production/operation) for MyHealth@EU routine operations.

For each matching candidate a single <subject/> element MUST be included within the control act wrapper.
In addition to the constraints defined in [IHE XCPD] the following conventions MUST be followed for <subject1/patient/> elements:

Element Name	Opt.	MyHealth@EU Usage Convention
Patient	R	For each matching candidate a single <patient/> element MUST be provided.
Patient/id	R	This element MUST contain the HL7-II-encoded Id of the patient that MUST be used for subsequent transactions to access the patent's medical data. The root designator MUST be present.
Patient/statusCode	R	MUST be "active".
Patient/patientPerson	R	Additional demographic data on a patient that matches the query. The encod ing of this data MUST follow the conventions as stated in [IHE XCPD]. See table below for a list of demographics that SHOULD be used for MyHealth@EU.

confidence level of 75%, then only 100% matches will be returned (see [Identity Management Specification] for details on ID traits matching and confidence levels).

Patient/subjectOf1/ queryMatchObservation	R	This element encodes the score of the match as an HL7 observation. It MUST be used as this: <hl7v3:queryMatchObservation classCode="OBS" mood-Code="EVN"> <hl7v3:code codeSystem="2.16.840.1.113883.1.11.19914"/> <hl7v3:value xsi:type="hl7v3:INT" value="***MATCH***"/> </hl7v3:queryMatchObservation>

Other elements MAY be provided within the result set by the sender but SHOULD be ignored by the receiver.

For a *FindIdentityByTraits* response only the following ID data MUST be provided as child elements of the <patientPerson/> element.

Identity Data	**Opt.**	**Usage Convention (if provided)**
asOtherIDs/id	O	This element SHOULD be only given if it provides further information on the scope and context of the used identification mechanism. This information SHOULD be suited to allow the HP to verify the claimed identity of the patient.
name	O	Both family name and given name SHOULD be provided. Note: This element is mandatory wrt. the HL7v3 schema. Therefore at least an empty instance MUST be included with the response.
birthTime	R+	MUST be provided as "YYYY[MM[DD[HHMM[SS[.S[S[S[S]]]]]]]][+/-ZZZZ]"
birthplace	O	SHOULD contain the country and city of birth
administrativeGenderCode	O	
addr	O	Only city and streetName SHOULD be provided
guardian	X	For the MyHealth@EU minors and dependent people will not be treated differently from others. This element MUST NOT be provided as no respective risk assessment has been done.

As specified in [IHE XCPD], the following status should be returned:

- AA (application accept) is returned in Acknowledgement.typeCode (transmission wrapper).
- OK (data found, no errors) is returned in QueryAck.queryResponseCode (control act wrapper).

3.1.2.1.9 Response Message (No Patient ID Discovered)

If the *MyHealth@EU Identification Service* provider does not find a matching patient identifier it SHOULD include a <reasonOf/> element with the response message:

```
<reasonOf typeCode="RSON">
  <detectedIssueEvent classCode="ALRT" moodCode="EVN">
    <code code="ActAdministrativeDetectedIssueCode" codeSystem="2.16.840.1.113883.5.4"/>
    <!— details on detected issue and proposed activity -->
  </detectedIssueEvent>
</reasonOf>
```

Depending on the reason for not providing a patient identifier, the codes and messages as defined below MUST be used[4].

[4] All codes using the coding system: codeSystem="1.3.6.1.4.1.19376.1.2.27.3 are to be used per XCPD error code definition.

Condition and proposed action	Reason Encoding
The service requestor tried an identification based on an ID only or did not provide enough data to univocally identify the patient. (WARNING) The HP SHOULD ask the patient for further demographics and re-issue the request. AA (application accept) is returned in Acknowledgement.typeCode (transmission wrapper). OK (data found, no errors) is returned in QueryAck.queryResponseCode (control act wrapper)	<triggerFor typeCode="TRIG"> <actOrderRequired classCode="ACT" moodCode="ENV"> <code code="AdditionalDemographicsRequested" codeSystem="1.3.6.1.4.1.12559.11.10.1.3.2.2.1"/ > </actOrderRequired> </triggerFor> If specific demographics are requested the respective code values of code system 1.3.6.1.4.1.19376.1.2.27.1 as specified in section 3.55.4.2.2.6 of [IHE XCPD] SHOULD be used. There may be as many triggerFor elements, each of them containing an ActOrderRequired element as needed to code the attributes which would increase the assurance of the match[5].
The service provider only allows for patient identification by national/shared ID (WARNING). The HP SHOULD ask the patient for a national (health care) identification card and re-issue the request using *Shared/national Patient Identifier Query and Feed mode.* AA (application accept) is returned in Acknowledgement.typeCode (transmission wrapper). AE (application error) is returned in QueryAck.queryResponseCode (control act wrapper)	<triggerFor typeCode="TRIG"> <actOrderRequired classCode="ACT" moodCode="ENV"> <code code="DemographicsQueryNotAllowed" codeSystem="1.3.6.1.4.1.12559.11.10.1.3.2.2.1"/ > </actOrderRequired> </triggerFor>
The service provider only allows for patient identification by national health card or EHIC. Queries based on demographics only are not supported (WARNING) The HP SHOULD ask the patient for a health care identification card and re-issue the request. AA (application accept) is returned in Acknowledgement.typeCode (transmission wrapper). AE (application error) is returned in QueryAck.queryResponseCode (control act wrapper)	<triggerFor typeCode="TRIG"> <actOrderRequired classCode="ACT" moodCode="ENV"> <code code="EHICDataRequested" codeSystem="1.3.6.1.4.1.12559.11.10.1.3.2.2.1"/ > </actOrderRequired> </triggerFor>
The service provider does not accept the query because responding MAY lead to a disclosure of private patient data (ERROR). The HP SHOULD limit the provided traits and re-issue the request.	<mitigatedBy typeCode="MITGT"> <detectedIssueManagement classCode="ACT" moodCode="ENV"> <code code="PrivacyViolation" codeSystem="1.3.6.1.4.1.12559.11.10.1.3.2.2.1"/ > </detectedIssueManagement> </mitigatedBy>

[5] See IHE ITI CP #535

AA (application accept) is returned in Acknowledgement.typeCode (transmission wrapper). AE (application error) is returned in QueryAck.queryResponseCode (control act wrapper)	
The requestor has insufficient rights to query for patient's identity data (ERROR). If access to the patient's medical data is required at the PoC this MUST be performed by a person with additional permissions. AA (application accept) is returned in Acknowledgement.typeCode (transmission wrapper). AE (application error) is returned in	<mitigatedBy typeCode="MITGT"> <detectedIssueManagement classCode="ACT" moodCode="ENV"> <code code="InsufficientRights" codeSystem="1.3.6.1.4.1.12559.11.10.1.3.2.2.1"/ > </detectedIssueManagement> </mitigatedBy>
Patient authentication MUST be piggybacked with patient identification. A respective identifier (e.g. GSS TAN) was not provided (ERROR) The HP at the PoC SHOULD ask the patient for a respective identifier and SHOULD re-issue the request. AA (application accept) is returned in Acknowledgement.typeCode (transmission wrapper). AE (application error) is returned in QueryAck.queryResponseCode (control act wrapper)	<mitigatedBy typeCode="MITGT"> <detectedIssueManagement classCode="ACT" moodCode="ENV"> <code code="PatientAuthenticationRequired" codeSystem="1.3.6.1.4.1.12559.11.10.1.3.2.2.1"/ > </detectedIssueManagement> </mitigatedBy>
The service provider did not find a match with the given minimum accuracy. (INFO) The service consumer SHOULD re-issue the request with a lower minimum confidence level. AA (application accept) is returned in Acknowledgement.typeCode (transmission wrapper). OK (data found) is returned in QueryAck.queryResponseCode (control act wrapper)	<mitigatedBy typeCode="MITGT"> <detectedIssueManagement classCode="ACT" moodCode="ENV"> <code code="AnswerNotAvailable" codeSystem="1.3.6.1.4.1.19376.1.2.27.3"/> </detectedIssueManagement> </mitigatedBy>
The identity traits provided by the service consumer are not supported by the service provider. (ERROR) The service consumer SHOULD re-issue the request with a different set of identity traits. AA (application accept) is returned in Acknowledgement.typeCode (transmission wrapper). AE (application error) is returned in QueryAck.queryResponseCode (control act wrapper)	<mitigatedBy typeCode="MITGT"> <detectedIssueManagement classCode="ACT" moodCode="ENV"> <code code="AnswerNotAvailable" codeSystem="1.3.6.1.4.1.19376.1.2.27.3"/> </detectedIssueManagement> </mitigatedBy>

The service consumer defined a confidence level that conflicts with the security policy of the service provider. (INFO) The service provider SHOULD respond only with the candidate matches that it is allowed to provide wrt. its security policy.	<mitigatedBy typeCode="MITGT"> <detectedIssueManagement classCode="ACT" moodCode="ENV"> <code code="PolicyViolation" codeSystem="1.3.6.1.4.1.12559.11.10.1.3.2.2.1"/> </detectedIssueManagement> </mitigatedBy>
AA (application accept) is returned in Acknowledgement.typeCode (transmission wrapper). AE (application error) is returned in QueryAck.queryResponseCode (control act wrapper)	

Additional information to describe the reason MUST be included in the acknowledgementDetail of the response.

AcknowledgementDetail.Code	The value MUST be a code listed in the tables of Appendix 6.4
AcknowledgementDetail.Text	The value is a text descirption of the error code
AcknowledgementDetail.Location	The location tag SHOULD be used to pass custom information from the NC-A and to the NC_B

3.1.2.1.10 Example Response Messages

The following sample message responds to a query with the patient identifier of a patient who matches the given identity traits. The match is unique and it is a full overlap with the given query.

```
<soapenv:Envelope>
<soapenv:Header> ... </soapenv:Header>
<soapenv:Body>
  <hl7v3:PRPA_IN201306UV02 xmlns:xsi="...">
    <hl7v3:id root="1.2.840.114350.1.13.999.238" extension="55789"/>
    <hl7v3:creationTime value="20100304110302"/>
    <hl7v3:interactionId root="2.16.840.1.113883.1.6" extension="PRPA_IN201306UV02"/>
    <hl7v3:processingCode code="P"/>
    <hl7v3:processingModeCode code="T"/>
    <hl7v3:acceptAckCode code="NE"/>
    <hl7v3:receiver typeCode="RCV">
      <hl7v3:device classCode="DEV" determinerCode="INSTANCE">
        <hl7v3:id root="1.2.840.114350.1.13.999.567"/>
      </hl7v3:device>
    </hl7v3:receiver>
    <hl7v3:sender typeCode="SND">
      <hl7v3:device classCode="DEV" determinerCode="INSTANCE">
        <hl7v3:id root="1.2.840.114350.1.13.999.234"/>
      </hl7v3:device>
    </hl7v3:sender>
    <hl7v3:controlActProcess classCode="CACT" moodCode="EVN">
      <hl7v3:code code="PRPA_TE201306UV02" codeSystem="2.16.840.1.113883.1.6"/>
      <hl7v3:subject typeCode="SUBJ">
        <hl7v3:registrationEvent classCode="REG" moodCode="EVN">
          <hl7v3:id nullFlavor="NA"/>
          <hl7v3:statusCode code="active"/>
          <hl7v3:subject1 typeCode="SBJ">
            <hl7v3:patient classCode="PAT">
              <!-- Identifier that MUST be used for subsequent requests -->
              <hl7v3:id  root="1.2.276.0.76.4.8" extension="1234567890"/>
              <hl7v3:statusCode code="active"/>
```

```
                                <hl7v3:patientPerson>
                                    <hl7v3:name/>
                                    <hl7v3:birthTime value="19680513"/>
                                </hl7v3:patientPerson>
                                <hl7v3:subjectOf1 typeCode="SBJ">
                                    <hl7v3:queryMatchObservation classCode="OBS" moodCode="EVN">
                                        <hl7v3:code codeSystem="2.16.840.1.113883.1.11.19914"/>
                                        <!-- Query score matching -->
                                        <hl7v3:value xsi:type="hl7v3:INT" value="100"/>
                                    </hl7v3:queryMatchObservation>
                                </hl7v3:subjectOf1>
                            </hl7v3:patient>
                        </hl7v3:subject1>
                        <hl7v3:custodian typeCode="CST">
                            <hl7v3:assignedEntity classCode="ASSIGNED">
                                <!-- Required element containing the homeCommunityId for the community responding to the
                                    request -->
                                        <hl7v3:id root="1.2.840.114350.1.13.99998.8734"/>
                                <!-- IHE Required element defining whether the responding community supports the QIL
                                    transaction for this patient,
                                    for MyHealth@EU the required value is "NotHealthDataLocator" -->
                                <hl7v3:code code="NotHealthDataLocator" codeSystem="1.3.6.1.4.1.19376.1.2.27.2"/>
                            </hl7v3:assignedEntity>
                        </hl7v3:custodian>
                    </hl7v3:registrationEvent>
                </hl7v3:subject>
                <hl7v3:queryAck>
                    <hl7v3:queryId root="1.2.840.114350.1.13.28.1.18.5.999" extension="18204"/>
                    <hl7v3:queryResponseCode code="OK"/>
                </hl7v3:queryAck>
            </hl7v3:controlActProcess>
        </soapenv:body>
    </soapenv:envelope>
```

The following sample message responds to a request that cannot be fulfilled because of insufficient traits.

```
<soapenv:Envelope>
<soapenv:Header> ... </soapenv:Header>
<soapenv:Body>
   <hl7v3:PRPA_IN201306UV02 xmlns:xsi="...">
      <hl7v3:id root="1.2.840.114350.1.13.999.238" extension="55789"/>
      <hl7v3:creationTime value="20100304110302"/>
      <hl7v3:interactionId root="2.16.840.1.113883.1.6" extension="PRPA_IN201306UV02"/>
      <hl7v3:processingCode code="P"/>
      <hl7v3:processingModeCode code="T"/>
      <hl7v3:acceptAckCode code="NE"/>
      <hl7v3:receiver typeCode="RCV">
         <hl7v3:device classCode="DEV" determinerCode="INSTANCE">
            <hl7v3:id root="1.2.840.114350.1.13.999.567"/>
         </hl7v3:device>
      </hl7v3:receiver>
      <hl7v3:sender typeCode="SND">
         <hl7v3:device classCode="DEV" determinerCode="INSTANCE">
            <hl7v3:id root="1.2.840.114350.1.13.999.234"/>
         </hl7v3:device>
      </hl7v3:sender>
      < hl7v3:acknowledgement>
         < hl7v3:typeCode code="AA"/>
         < hl7v3:targetMessage>
                  < hl7v3:id extension="1419510353" root="d3b67e00-1214-4ef7-9c02-310e2df08f4c"/>
```

```
</ hl7v3:targetMessage>
< hl7v3:acknowledgementDetail>
  < hl7v3:code code="ERROR_PI_NO_MATCH"/>
  < hl7v3:text> he identification and authentication used in the Country of Treatment (Country B) are  not correctly
  provided or does not match any existing patient.
  </ hl7v3:text>
  < hl7v3:location>additional custom information from country A</ hl7v3:location>
</ hl7v3:acknowledgementDetail>
</ hl7v3:acknowledgement>
<hl7v3:controlActProcess classCode="CACT" moodCode="EVN">
    <hl7v3:code code="PRPA_TE201306UV02" codeSystem="2.16.840.1.113883.1.6"/>
    <!-- Used to indicate that more attributes are required -->
    <hl7v3:reasonOf typeCode="RSON">
        <hl7v3:detectedIssueEvent classCode="ALRT" moodCode="EVN">
            <hl7v3:code code="ActAdministrativeDetectedIssueCode" codeSystem="2.16.840.1.113883.5.4"/>
                                        <hl7v3:triggerFor typeCode="TRIG">
                <hl7v3:actOrderRequired classCode="ACT" moodCode="ENV">
                  <hl7v3:code code="AdditionalDemographicsRequested"
                codeSystem="1.3.6.1.4.1.12559.11.10.1.3.2.2.1.1"/>
                  </hl7v3:actOrderRequired>
              </hl7v3:triggerFor>
          </hl7v3:detectedIssueEvent>
      </hl7v3:reasonOf>
      <hl7v3:queryAck>
          <hl7v3:queryId root="1.2.840.114350.1.13.28.1.18.5.999" extension="18204"/>
          <hl7v3:queryResponseCode code="OK"/>
    </hl7v3:queryAck>
  </hl7v3:controlActProcess>
</hl7v3:PRPA_IN201306UV02>
</soapenv:body>
</soapenv:envelope>
```

3.1.2.2 Security Audit Considerations

Both the *MyHealth@EU Identification Service* provider and consumer write an audit trail entry according to the ID Mapping audit schema. The following table defines which categories MUST be filled (R), which MAY be filled (O) and which categories MUST NOT be used (X).

MyHealth@EU Instance	Opt.	Description
Event	R	Audited event
Human Requestor	R	HP who triggered the event
Source Gateway	R	Service consumer node address at the country of care
Target Gateway	R	Service provider node address at the country of affiliation
Mapping Service	R / X	Service that provided the mapping. MUST be filled by the service provider. MUST NOT be filled by the service consumer.
Audit Source	R	Legal entity that ensures the uniqueness of the identifiers that are used to identify active participants
Patient Source	R	Patient whose identifier was discovered or mapped
Patient Target	R	Result of the mapping operation
Error Message	O	Only used in case that the transaction was not completed successfully

Table 2: MyHealth@EU Patient Identification Service Audit Message Categories

3.1.2.3 Protocol Requirements

The *MyHealth@EU Patient Identification Service* FindIdentityByTraits request and response messages will be transmitted using synchronous Web Services Exchange, according to the requirements specified in section 4.3 of this document.

Port types and bindings MUST be used as defined in the WSDL given in section 6.4.1 of this document. Acc. to this the MyHealth@EU FindIdentityByTraits operation's request and response data MUST be contained within the message body as follows:

MyHealth@EU Patient Identification Service	**Message Body**
FindIdentityByTraits request	PRPA_IN201305UV02_Message (see section 6.4.1)
FindIdentityByTraits response	PRPA_IN201306UV02_Message (see section 6.4.1)

The request message MUST be protected by the service consumer (NCP-B) according to the MyHealth@EU message security considerations as defined in section 4.3.5.2 of this document. The response message MUST be protected by the service provider (NCP-A) according to the MyHealth@EU message security considerations as defined in section 4.3.5.2 of this document.

3.1.3 MyHealth@EU EUDI Wallet for findIdentityByTraits()

The findIdentityByTraits() operation defined in section 3.1.2.1 requires the NCP-B to supply patient identity traits such as LivingSubjectId, LivingSubjectName, and LivingSubjectBirthTime in order to drive the XCPD Cross-Gateway Patient Discovery transaction toward NCP-A.
Traditionally these traits are entered manually by the Healthcare Professional (HP) at the NCP-B Front-End Portal.
CP090 introduces an optional, alternative trait-sourcing mechanism: the HP may trigger a EUDI Wallet presentation request, which allows the patient to present their Person Identification Data (PID) credential — issued by their home member state — directly from their EUDI Wallet app.
The verified PID claims are then mapped onto the findIdentityByTraits() parameter set without HP transcription.
This mechanism is strictly optional at NCP-B level and introduces no changes to the findIdentityByTraits() operation itself, to its SOAP message format, or to any NCP-A component. All normative constraints defined in section 3.1.2.1 remain fully applicable.

3.1.4 MyHealth@EU Patient Service

The *MyHealth@EU Patient Service* is used to share an identified patient's medical summary between the patient's country of affiliation and the country of care. Both countries are represented by their respective NCPs.



Figure 9 - Patient Service Interface

The implementation of the MyHealth@EU Patient Service is based on the following standards:

- ebRIM: OASIS/ebXML Registry Information Model v3.0 [OASIS ebRIM

3.0]

- ebRS: OASIS/ebXML Registry Services Specifications v3.0[6] [OASIS ebRS 3.0]
- XOP: XML-binary Optimized Packaging [W3C XOP] and is based on the following IHE profile:
- XCA: IHE Cross-Community Access [IHE XCA]

For discovery and localisation of the Patient Service instance that is responsible for providing access to the identified patient's data see section 2.1.4 of this document.

3.1.4.1 List() Operation

The *MyHealth@EU Patient Service* list() operation is implemented as IHE XCA Cross-Community Access transaction. It is fully compliant with the ebRS 3.0 standard. The *MyHealth@EU Patient Service* list() operation includes the documents listed in the response meta-data, just like they would have been included in Cross-Gateway Fetch (SOAP 1.2 MTOP with XOP encoding attachments).

3.1.4.1.1 Request Message

The list() request is initiated by an HP in the country of care for retrieving the patient summary of an identified patient. The respective request message builds upon the IHE Cross-Community Access request message.

The <AdhocQueryRequest/> element that encapsulates the query parameters MUST be used as follows for MyHealth@EU:

Element Name	MyHealth@EU Usage Convention
ResponseOption/@returnComposedObjects	MUST be "true"
ResponseOption/@returnType	MUST be "LeafClassWithRepositoryItem" (XCF)
AdhocQuery	Container for holding the ebML stored query arguments. All arguments MUST be encoded as query slots (see table below).
AdhocQuery@id	MUST be "urn:uuid:f2072993-9478-41df-a603-8f016706efe8" which indicates a Fetch (which is an adaption of the findDocuments Query as defined in ITI TF-2a:3.18.1)

Only synchronous web services exchange MUST be used. The *XDS Affinity Domain Option* only applies to the national environment. Therefore it MUST NOT be used for NCP-2-NCP message exchange.

Stored query argument slots MUST be defined for the patient identifier and the document class code. The document format code and the document type code MAY be given. Other argument slots than the ones listed below MUST be ignored by the service provider and SHOULD NOT be issued by the service consumer.

Slot Name	Opt	Slot Value
$XDSDocumentEntryPatientId	R	Equals to the patient identifier that was provided by the *MyHealth@EU Identification Service* (encoded as HL7 v3 II data type)
$XDSDocumentEntryStatus	R	Only approved documents MUST be returned: 'urn:oasis:names:tc:ebxml-regrep: StatusType:Approved'

[6] The integration of ebRS and MTOM as used by MyHealth@EU is not compatible with the current version of OASIS ebRS. Support for MTOM will be part of the forthcoming ebRS v4.0.

$XDSDocumentEntryClassCode	R	Patient summary LOINC code ("60591-5") coded according to specification in ITI TF-2a: 3.18.4.1.2.3.4 Coding of Code/Code-Scheme. As classification scheme 2.16.840.1.113883.6.1 MUST be used: '60591-5^^2.16.840.1.113883.6.1'
$XDSDocumentEntryTypeCode	O	Patient summary LOINC code ("60591-5") coded according to specification in ITI TF-2a: 3.18.4.1.2.3.4 Coding of Code/Code-Scheme. As classification scheme 2.16.840.1.113883.6.1 MUST be used:
$XDSDocumentEntryFormatCode	O	Format qualifier as defined in [CDA templates]; see table below for details on applying these codes to the retrieval of a patient's medical summary. Only encodings of the patient summary that comply to the requested format code will be returend by the service provider. If this stored query slot is omitted the service provider MUST deliver all available encodings[7].

For the document format only the format codes defined in [CDA templates] and listed in the following table MUST be used.

Document Format	Format Code	Document content
MyHealth@EU pivot coded Patient Summary	urn:epsos:ps:ps:2010	HL7 CDA document acc. [CDA templates]. The patient's country of affiliation MUST be able to provide the patient's summarised medical data in this format.
PDF/A source coded document	urn:ihe:iti:xds-sd:pdf:2008	CDA-enveloped PDF/A encoding of the original document without any semantic transformation of a patient summary as source coded PDF with a CDA header per IHE XDS-SD. The patient's country of affiliation SHOULD be able to provide the patient's summarised medical data in this format.

3.1.4.1.2 Example Request Message

The following excerpt from an *MyHealth@EU Patient Service* list() request message shows a cross-NCP query request that contains argument slots for retrieving the patient summary (LOINC code 60591-5) of an identified patient (patient identifier 90378912821). In this example the service consumer does not specify the requested encoding. Therefore the service provider MUST deliver all available encodings (e.g. MyHealth@EU pivot and source coded document).

```
<soapenv:Envelope xmlns:soapenv=" http://www.w3.org/2003/05/soap-envelope" ... >
<soapenv:Header> ... </soapenv:Header>
<soapenv:Body>
   <query:AdhocQueryRequest>
      <query:ResponseOption returnComposedObjects="true"
                                                    returnType="LeafClassWithRepositoryItem"/>
      <rim:AdhocQuery id="urn:uuid:14d4debf-8f97-4251-9a74-a90016b0af0d">
```

[7] Acc. to [PS Functional requirements] countries MAY provide patient summary data only in MyHealth@EU pivot coded format. A query where the format code is omitted will in these cases provide the same result as a query for the MyHealth@EU pivot coded document format only.

```
            <rim:Slot name="$XDSDocumentEntryPatientId">
                <rim:ValueList>
                    <rim:Value>'90378912821^^^&amp;amp;1.3.6.1.4.1.21367.2005.3.7&amp;amp;ISO'
                    </rim:Value>
                </rim:ValueList>
            </rim:Slot>
            <rim:Slot name="$XDSDocumentEntryStatus">
                <rim:ValueList>
                    <rim:Value>('urn:oasis:names:tc:ebxml-regrep:StatusType:Approved')
                    </rim:Value>
                </rim:ValueList>
            </rim:Slot>
            <rim:Slot name="$XDSDocumentEntryClassCode">
                <rim:ValueList>
                    <rim:Value>('60591-5^^2.16.840.1.113883.6.1')</rim:Value>
                </rim:ValueList>
            </rim:Slot>
            <!-- Include associations whose sourceObject and targeObject attributes
                 reference ExtrinsicObjects returned -->
        </rim:AdhocQuery>
    </query:AdhocQueryRequest>
</soapenv:Body>
</soapenv:Envelope>
```

3.1.4.1.3 Expected Actions

The *MyHealth@EU Patient Service* provider shall respond to a ListRequest message with the ListResponse message containing

- the identified patient's patient summary document(s) together with a status notification (full success scenario, see section 3.3.1.4) or
- an error message (no patient summary provided, see section 3.3.1.5).

The *MyHealth@EU Patient Service* provider MUST verify that the requesting service user has sufficient rights to access the full patient summary of the identified patient.
In case of an error that relates to the transmission of the request or the processing of the MyHealth@EU security token, the *MyHealth@EU Patient Service* provider MUST respond with a fault message according to section 4.4 of this document.

3.1.4.1.4 Response Message (Full Success Scenario)

Depending on the requested format code the MyHealth@EU list() response contains the MyHealth@EU pivot encoded patient summary document, the PDF/A encoded patient summary document or both documents of the identified patient. The respective message builds upon the IHE XCA Cross-Community Access response and Cross-Gateway Fetch Response messages.
The fields defined for the MyHealth@EU ListResponse message MUST be used as follows:

Element Name	MyHealth@EU Usage Convention
query:AdhocQueryResponse	Response message acc to IHE XCA Cross-Community Access response message [IHE XCA]
@status	For the full success scenario the response status MUST be set to "urn:oasis:names:tc:ebxml-regrep:ResponseStatusType:Success" or "urn:ihe:iti:2007:ResponseStatusType:PartialSuccess" (for details see table below)
../rs:RegistryErrorList	In case that a warning is given by the service provider, this element holds the respective warning codes and messages. It must be used acc. to section 4.1.13 of [IHE ITI TF-3].
../rim:RegistryObjectList	This element MUST be provided for the full success scenario. It MUST at least contain one child <rim:ExtrinsicObject/> element.

../../rim:ExtrinsicObject	For each encoding of the patient summary a <rim:ExtrinsicObject/> element MUST be provided. Each <rim:ExtrinsicObject/> element is described and classified by metadata acc. to Table 3 below.
../../../rimext:Document	This element MUST appear as the last element child of an <rim:ExtrinsicObject/> element. It may appear zero or one times. This element contains the base 64 encoded content of the document. The document contents are associated with the DocumentEntry (ExtrinsicObject) metadata by the fact that it is nested inside it within the XML. The base64 encoded document content MAY be encrypted. How encryption is applied and how the encryption key is negotiated should be subject to an additional specification on advanced security safeguards.

Each provided patient summary encoding (MyHealth@EU pivot and/or source coded PDF) MUST be further classified by metadata. The following table lists the usage conventions that MUST be followed for the MyHealth@EU Patient Service response message. If not stated otherwise the classification schemes as defined in section 4.3.1.2 of [IHE ITI TF-3] MUST be used. If no restrictions on metadata values are given, the metadata elements MUST be used as per [IHE XCA].

Metadata (ebRIM names)	**Binding**	**Opt.**	**MyHealth@EU usage convention**
status	Attribute	R	MUST be "urn:oasis:names:tc:ebxml-regrep:StatusType:Approved"
mimeType	Attribute	R	MUST be "text/xml" for both MyHealth@EU pivot CDA and CDA-
Name	Main	R	MUST be "Patient Summary".
Description	Main	O	MAY be empty. MAY be ignored by the
VersionInfo	Main	R	MUST be "1.1"
creationTime	rim:Slot	O	MAY be omitted by the service provider and MAY be ig- nored by the service consumer. If given, the value MUST be encoded as HL7 v2 Date Time "YYYY[MM[DD[hh[mm[ss]]]]]"
hash	rim:Slot	O	SHOULD be omitted by the service provider and MUST NOT be processed by the service consumer.
languageCode	rim:Slot	O	SHOULD be omitted by the service provider and MUST NOT be processed by the service consumer.
repositoryUniqueId	rim:Slot	O	COULD be omitted by the service provider and MAY be processed by the service consumer in an IHE-compatible NI-scenario.
serviceStartTime serviceEndTime	rim:Slot	O	SHOULD be omitted by the service provider and MUST NOT be processed by the service consumer.
size	rim:Slot	O	SHOULD be omitted by the service provider and MUST NOT be processed by

sourcePatientId	rim:Slot	R	MUST contain the same value as XDSDocumentEntry PatientID (see below).
sourcePatientInfo	rim:Slot	X	MUST NOT be used. Future versions of MyHealth@EU MAY define different protection levels for metadata and documents. Therefore all metadata elements that might carry medical or social information MUST be omitted.
classCode	Classification	R	Patient summary LOINC code ("60591-5"). As classifica- tion scheme "urn:oid:2.16.840.1.113883.6.1" MUST be used
eventCodeList	Classification	X	MUST NOT be used. Future versions of MyHealth@EU MAY define different protection levels for metadata and docu- ments. Therefore all metadata elements that might carry medical or social information MUST be omitted.
author	Classification	X	MUST NOT be used. Future versions of MyHealth@EU MAY define different protection levels for metadata and documents. Therefore all metadata elements that might carry medical or social information MUST be omitted.
confidentialityCode	Classification	R	MUST be provided for XCA compatibility but MAY be ignored by the service consumer. Value SHOULD be set to "N", as long as the Minimal Metadata Profile is not published.
formatCode	Classification	R	MUST be "urn:epSOS:ps:ps:2010" for MyHealth@EU pivot CDA and "urn:ihe:iti:xds-sd:pdf:2008" for MyHealth@EU source coded PDF (see [CDA templates]).
healthcareFacilityTypeCode	Classification	R	MUST be provided for XCA compatibility and correct ad- dressing. Value MUST be set to ISO 3166-1 alpha-2 coun- try code of the addressed PN.
practiceSettingCode	Classification	R	MUST be provided for XCA compatibility. Value MUST be set to "Not Used" in order to protect private patient information.
XDSDocumentEntry.uniqueId	ExternalIdentifier	R	MUST hold the OID of the document. The document unique id value MUST be the same as the value of the document's <ClinicalDocument/id> CDA header element.
XDSDocumentEntry.patientId	ExternalIdentifier	R	MUST hold the patient identifier. The service consumer MUST verify that this id matches the patient Id that was discovered by the MyHealth@EU Identification Service.

Table 3: MyHealth@EU Patient Summary Metadata

Other metadata than the ones listed above SHOULD NOT be provided by the service provider and MUST NOT be processed by the service consumer.
By definition only a single patient summary is provided per patient [PS Functional

requirements]. If two documents are provided in response to a Patient Service list request, these MUST be different encodings of the same patient's medical summary data (MyHealth@EU pivot coded and PDF/A source coded).
If document relationships are defined, an ebRIM association MUST be used for declaring the MyHealth@EU pivot coded document as a transformation of the source coded document. As classification scheme urn:uuid:abd807a3-4432-4053-87b4-fd82c643d1f3 MUST be used per IHE XCF. "MyHealth@EU pivot" is defined as the only valid code value for the transformation:

```
<rim:Association id="id of the association" associationType="urn:ihe:iti:2007:AssociationType:XFRM" sourceObject="UUID of
    the source coded document" targetObject="UUID of the MyHealth@EU pivot document"
    objectType="urn:oasis:names:tc:ebxml-regrep:ObjectType:RegistryObject:Classification">
    <rim:Classification
    id="id of the classification"
   classificationScheme="urn:uuid:abd807a3-4432-4053-87b4-fd82c643d1f3" classifiedObject="id of the association"
    objectType="urn:oasis:names:tc:ebxml-regrep:ObjectType:RegistryObject:Classification" nodeRepresentation="epSOS
    pivot">
        <rim:Slot name="codingScheme">
            <rim:ValueList>
                <rim:Value>MyHealth@EU translation types</rim:Value>
            </rim:ValueList>
        </rim:Slot>
        <rim:Name>
            <rim:LocalizedString value="Translation into MyHealth@EU pivot format"/>
        </rim:Name>
    </rim:Classification>
</rim:Association>
```

Metadata (ebRIM names)	Binding	Opt	MyHealth@EU usage convention
associationType	Attribute	R	MUST be "urn:ihe:iti:2007:AssociationType:XFRM"
sourceObject	Attribute	R	MUST be UUID of the source coded document
targetObject	Attribute	R	MUST be UUID of the MyHealth@EU pivot document
Association Documentation	Main	O	MAY be empty. MAY be ignored by the service consumer. MAY contain a description of the Association according to the Association Documentation object as used by IHE.
classificationSheme	classification	O	MUST be "urn:uuid:abd807a3-4432-4053-87b4- fd82c643d1f3"
classifiedObject	Attribute	O	MUST be I.D. of the association
nodeRepresentation	Attribute	O	MUST be "MyHealth@EU pivot"
codingScheme	rim:Slot	O	MUST be "MyHealth@EU translation types" and "Translation into MyHealth@EU pivot format"

If a warning is to be transmitted to the HP (see section 3.1.4) the ebXML Registry Error mechanism MUST be used with a syntax as defined in section 3.43.5 of [IHE ITI TF-2]. As the location of the warning is implied, the respective location attribute SHOULD be empty. The table lists from Appendix 6.4, defined MyHealth@EU warning codes:

3.1.4.1.5 Response Message (No Patient Summary Provided)

If the *MyHealth@EU Patient Service* provider is unable to respond with the patient's summarised medical data in the requested encoding it MUST respond with a ListResponse message that only contains a <AdhocQueryResponse/RegistryResponse>element.

For a full list of error messages defined for IHE X* see table 4.1-11 in [IHE ITI TF-3]. The table from Appendix 6.4 lists the additional, MyHealth@EU-specific response status types and error/warning/info codes to be used within the <RegistryErrorList> element.

3.1.4.1.6 Example Response Message

The following message is a possible response to the sample request message given in section 3.3.1.2. The patient's country of affiliation responds with both encodings.

```
<soapenv:Envelope>
    <soapenv:Header>....</soapenv:Header>
    <soapenv:Body>
     <query:AdhocQueryResponse
                status="urn:oasis:names:tc:ebxml-regrep:ResponseStatusType:Success">
                                                <rim:RegistryObjectList>
                                    <!— MyHealth@EU pivot CDA patient summary document -->
            <rimext:ExtrinsicObject
               id="urn:uuid:fbf2ea29-3aa3-4bc5-9187-01d7b6b0f481"
               lid="urn:uuid:fbf2ea29-3aa3-4bc5-9187-01d7b6b0f481" objectType="urn:uuid:7edca82f-054d-47f2-a032-
               9b2a5b5186c1" status="urn:oasis:names:tc:ebxml-regrep:StatusType:Approved" mimeType="text/xml">

               <!-- These attributes are required by XCA but not used by MyHealth@EU.
                     They will be ignored by the MyHealth@EU service consumer (NCP-B) -->
               <rim:Slot name="creationTime">
                     <rim:ValueList>
                           <rim:Value>20100524</rim:Value>
                     </rim:ValueList>
               </rim:Slot>
               <rim:Slot name="languageCode">
                     <rim:ValueList>
                           <rim:Value>en-us</rim:Value>
                     </rim:ValueList>
               </rim:Slot>
               <!-- set to same value as Patient ID (required by XCA) -->
               <rim:Slot name="sourcePatientId">
                     <rim:ValueList>
                           <rim:Value>90378912821^^^&amp;1.3.6.1.4.1.21367.2005.3.7&amp;ISO</rim:Value>
                     </rim:ValueList>
               </rim:Slot>
               <rim:Name>
                     <rim:LocalizedString xml:lang="en" charset="UTF-8"
                                                            value="Patient Summary"/>
               </rim:Name>
               <rim:Description/>
               <rim:VersionInfo versionName="1.1"/>
               <!-- HealthcareFacilityType Code -->
               <rim:Classification
                     id="urn:uuid:5c678da8-6ffa-4a85-90f6-cb2f914d482f" lid="urn:uuid:5c678da8-6ffa-4a85-90f6-
                     cb2f914d482f" objectType="urn:oasis:names:tc:ebxml-
                                 regrep:ObjectType:RegistryObject:Classification"
                     classificationScheme="urn:uuid:f33fb8ac-18af-42cc-ae0e-ed0b0bdb91e1"
                     classifiedObject="urn:uuid:fbf2ea29-3aa3-4bc5-9187-01d7b6b0f481" nodeRepresentation="Not Used">
                     <rim:Slot name="codingScheme">
                     <rim:ValueList>
                     <rim:Value>
                                 MyHealth@EU Healthcare Facility Type Codes-Not Used
                     </rim:Value>
                     </rim:ValueList>
                     </rim:Slot>
                     <rim:Name>
                           <rim:LocalizedString xml:lang="en" charset="UTF-8" value="Not Used"/>
                     </rim:Name>
               </rim:Classification>
               <!-- PracticeSetting Code -->
               <rim:Classification
                     id="urn:uuid:b01599e3-79a6-4322-b5fc-f32ada9ee7f4"
                     lid="urn:uuid:b01599e3-79a6-4322-b5fc-f32ada9ee7f4" objectType="urn:oasis:names:tc:ebxml-
regrep:ObjectType:RegistryObject:Classification" classificationScheme="urn:uuid:cccf5598-8b07-4b77-a05e-ae952c785ead"
                     classifiedObject="urn:uuid:fbf2ea29-3aa3-4bc5-9187-01d7b6b0f481" nodeRepresentation="Not Used">
                     <rim:Slot name="codingScheme">
                           <rim:ValueList>
                                 <rim:Value>
                                  MyHealth@EU Practice Setting Codes-Not Used
                                 </rim:Value>
                           </rim:ValueList>
```

```
                    </rim:Slot>
                    <rim:Name>
                        <rim:LocalizedString xml:lang="en" charset="UTF-8" value="Not Used"/>
                    </rim:Name>
                </rim:Classification>
                <!-- Confidentiality Code -->
                <rim:Classification
                    id="urn:uuid:d0dc74b9-f013-4639-b9c2-fac2420af0dd"
                    lid="urn:uuid:d0dc74b9-f013-4639-b9c2-fac2420af0dd" objectType="urn:oasis:names:tc:ebxml-
regrep:ObjectType:RegistryObject:Classification" classificationScheme="urn:uuid:f4f85eac-e6cb-4883-b524-f2705394840f"
                    classifiedObject="urn:uuid:fbf2ea29-3aa3-4bc5-9187-01d7b6b0f481" nodeRepresentation="Not Used">
                    <rim:Slot name="codingScheme">
                        <rim:ValueList>
                            <rim:Value>
                                MyHealth@EU Confidentiality Codes-Not Used
                            </rim:Value>
                        </rim:ValueList>
                    </rim:Slot>
                    <rim:Name>
                        <rim:LocalizedString xml:lang="en" charset="UTF-8" value="Not Used"/>
                    </rim:Name>
                </rim:Classification>
                <!-- End of attributes not used by MyHealth@EU -->
                <!-- Class Code -  (60591-560591-5) -->
                <rim:Classification
                    id="urn:uuid:c33ca26a-29b4-45be-a9b9-de60adca4c64" lid="urn:uuid:c33ca26a-29b4-45be-a9b9-
                    de60adca4c64" objectType="urn:oasis:names:tc:ebxml-
regrep:ObjectType:RegistryObject:Classification"
                    classificationScheme="urn:uuid:41a5887f-8865-4c09-adf7-e362475b143a"
                    classifiedObject="urn:uuid:fbf2ea29-3aa3-4bc5-9187-01d7b6b0f481" nodeRepresentation="60591-5">
                    <rim:Slot name="codingScheme">
                        <rim:ValueList>
                            <rim:Value>2.16.840.1.113883.6.1</rim:Value>
                        </rim:ValueList>
                    </rim:Slot>
                    <rim:Name>
                        <rim:LocalizedString xml:lang="en" charset="UTF-8"
                            value="Patient Summary"/>
                    </rim:Name>
                </rim:Classification>
                <!-- Type Code -  (60591-5) -->
                <rim:Classification
                    id="urn:uuid:87a7cfc2-a956-4d6e-af30-c7e78809c95f"
                    lid="urn:uuid:87a7cfc2-a956-4d6e-af30-c7e78809c95f" objectType="urn:oasis:names:tc:ebxml-
regrep:ObjectType:RegistryObject:Classification" classificationScheme="urn:uuid:f0306f51-975f-434e-a61c-c59651d33983"
                    classifiedObject="urn:uuid:fbf2ea29-3aa3-4bc5-9187-01d7b6b0f481" nodeRepresentation="60591-5">
                    <rim:Slot name="codingScheme">
                        <rim:ValueList>
                            <rim:Value>2.16.840.1.113883.6.1</rim:Value>
                        </rim:ValueList>
                    </rim:Slot>
                    <rim:Name>
                        <rim:LocalizedString xml:lang="en" charset="UTF-8" value="Patient Summary"/>
                    </rim:Name>
                </rim:Classification>
                <!-- Format Code -->
                <rim:Classification
                    id="urn:uuid:ae68bdf8-4f32-4829-8313-2dd39ea3ab2d"
                    lid="urn:uuid:ae68bdf8-4f32-4829-8313-2dd39ea3ab2d" objectType="urn:oasis:names:tc:ebxml-
regrep:ObjectType:RegistryObject:Classification"
                    classificationScheme="urn:uuid:a09d5840-386c-46f2-b5ad-9c3699a4309d"
                    classifiedObject="urn:uuid:fbf2ea29-3aa3-4bc5-9187-01d7b6b0f481"
                    nodeRepresentation="MyHealth@EU coded summary">
                  <rim:Slot name="codingScheme">
                    <rim:ValueList>
                        <rim:Value>MyHealth@EU formatCodes</rim:Value>
                    </rim:ValueList>
                    </rim:Slot>
                    <rim:Name>
                        <rim:LocalizedString xml:lang="en" charset="UTF-8" value="MyHealth@EU Coded Summary"/>
                    </rim:Name>
                </rim:Classification>

                <!-- Patient ID -->
                <rim:ExternalIdentifier
                    id="urn:uuid:982f1551-5901-4bc5-8870-801181941817"
                    lid="urn:uuid:982f1551-5901-4bc5-8870-801181941817" objectType="urn:oasis:names:tc:ebxml-
```

```
regrep:ObjectType:RegistryObject:ExternalIdentifier" identificationScheme="urn:uuid:58a6f841-87b3-4a3e-92fd-a8ffeff98427"
                        value="90378912821^^^&amp;1.3.6.1.4.1.21367.2005.3.7&amp;ISO" registryObject="urn:uuid:fbf2ea29-
                        3aa3-4bc5-9187-01d7b6b0f481">
                        <rim:Name>
                            <rim:LocalizedString xml:lang="en-us" charset="UTF-8"
                                                        value="XDSDocumentEntry.patientId"/>
                        </rim:Name>
                    </rim:ExternalIdentifier>
                    <!-- Unique ID -->
                    <rim:ExternalIdentifier
                        id="urn:uuid:c67e3a92-5300-448d-9af2-0a37e9f129bf"
                        lid="urn:uuid:c67e3a92-5300-448d-9af2-0a37e9f129bf" objectType="urn:oasis:names:tc:ebxml-
regrep:ObjectType:RegistryObject:ExternalIdentifier" identificationScheme="urn:uuid:2e82c1f6-a085-4c72-9da3-8640a32e42ab"
                        value="1.42.20100103225206.3.3"
                        registryObject="urn:uuid:fbf2ea29-3aa3-4bc5-9187-01d7b6b0f481">
                        <rim:Name>
                            <rim:LocalizedString xml:lang="en-us" charset="UTF-8"
                                                        value="XDSDocumentEntry.uniqueId"/>
                        </rim:Name>
                    </rim:ExternalIdentifier><rimext:Document>
                        UjBsR09EbGhjZ0dTQUxNQUFBUUNBRU1tQ1p0dU1GUXhEUzhi.....
                    </rimext:Document>
                </rimext:ExtrinsicObject>
                 <!—MyHealth@EU source coded PDF patient summary document -->
                <rimext:ExtrinsicObject
                    id="urn:uuid:a1c7a9ac-83aa-4eaf-b5e3-d355b57a5016" lid="urn:uuid:a1c7a9ac-83aa-4eaf-b5e3-d355b57a5016"
                    objectType="urn:uuid:7edca82f-054d-47f2-a032-9b2a5b5186c1" status="urn:oasis:names:tc:ebxml-
                    regrep:StatusType:Approved" mimeType="text/xml">

                    <!-- These attributes are required by XCA but not used by MyHealth@EU.
                            They will be ignored by the MyHealth@EU service consumer (NCP-B) -->
                    <rim:Slot name="creationTime">
                        <rim:ValueList>
                            <rim:Value>20100524</rim:Value>
                        </rim:ValueList>
                    </rim:Slot>
                    <rim:Slot name="languageCode">
                        <rim:ValueList>
                            <rim:Value>en-us</rim:Value>
                        </rim:ValueList>
                    </rim:Slot>
                    <!-- set to same value as Patient ID (required by XCA) -->
                    <rim:Slot name="sourcePatientId">
                        <rim:ValueList>
                            <rim:Value>
                                90378912821^^^&amp;1.3.6.1.4.1.21367.2005.3.7&amp;ISO
                            </rim:Value>
                        </rim:ValueList>
                    </rim:Slot>
                    <rim:Name/>
                    <rim:Description/>
                    <rim:VersionInfo versionName="1.1"/>
                    <!-- HealthcareFacilityType Code -->
                    <rim:Classification
                        id="urn:uuid:7dda3d1e-8d96-4fee-b691-f1810d44bc8d"
                        lid="urn:uuid:7dda3d1e-8d96-4fee-b691-f1810d44bc8d" objectType="urn:oasis:names:tc:ebxml-
regrep:ObjectType:RegistryObject:Classification" classificationScheme="urn:uuid:f33fb8ac-18af-42cc-ae0e-ed0b0bdb91e1"
                        classifiedObject="urn:uuid:a1c7a9ac-83aa-4eaf-b5e3-d355b57a5016" nodeRepresentation="Not Used">
                        <rim:Slot name="codingScheme">
                        <rim:ValueList>
                            <rim:Value>
                                    MyHealth@EU Healthcare Facility Type Codes-Not Used
                            </rim:Value>
                        </rim:ValueList>
                        </rim:Slot>
                        <rim:Name>
                            <rim:LocalizedString xml:lang="en" charset="UTF-8" value="Not Used"/>
                        </rim:Name>
                    </rim:Classification>
                    <!-- PracticeSetting Code -->
                    <rim:Classification
                        id="urn:uuid:89a73ab3-344a-4098-b827-aca8dea078ef" lid="urn:uuid:89a73ab3-344a-4098-b827-
                        aca8dea078ef" objectType="urn:oasis:names:tc:ebxml-
regrep:ObjectType:RegistryObject:Classification"
                        classificationScheme="urn:uuid:cccf5598-8b07-4b77-a05e-ae952c785ead"
                        classifiedObject="urn:uuid:a1c7a9ac-83aa-4eaf-b5e3-d355b57a5016" nodeRepresentation="Not Used">
                        <rim:Slot name="codingScheme">
```

```
                        <rim:ValueList>
                            <rim:Value>
                                    MyHealth@EU Practice Setting Codes-Not Used
                            </rim:Value>
                        </rim:ValueList>
                        </rim:Slot>
                        <rim:Name>
                            <rim:LocalizedString xml:lang="en" charset="UTF-8"
                                value="Not Used"/>
                        </rim:Name>
                    </rim:Classification>
                    <!-- Confidentiality Code -->
                    <rim:Classification
                        id="urn:uuid:fa176711-e83a-4fb2-95a3-a4810b0351fa" lid="urn:uuid:fa176711-e83a-4fb2-95a3-
                        a4810b0351fa" objectType="urn:oasis:names:tc:ebxml-
regrep:ObjectType:RegistryObject:Classification" classificationScheme="urn:uuid:f4f85eac-e6cb-4883-b524-f2705394840f"
                        classifiedObject="urn:uuid:a1c7a9ac-83aa-4eaf-b5e3-d355b57a5016" nodeRepresentation="Not Used">
                        <rim:Slot name="codingScheme">
                        <rim:ValueList>
                            <rim:Value>
                                    MyHealth@EU Confidentiality Codes-Not Used
                            </rim:Value>
                        </rim:ValueList>
                        </rim:Slot>
                        <rim:Name>
                            <rim:LocalizedString xml:lang="en" charset="UTF-8" value="Not Used"/>
                        </rim:Name>
                    </rim:Classification>
                    <!-- End of attributes not used by MyHealth@EU -->
                    <!-- Class Code - Patient Summary (60591-5) -->
                    <rim:Classification
                        id="urn:uuid:8a07ab13-1685-452f-9363-c89a37d9eb5b"
                        lid="urn:uuid:8a07ab13-1685-452f-9363-c89a37d9eb5b" objectType="urn:oasis:names:tc:ebxml-
regrep:ObjectType:RegistryObject:Classification"
                        classificationScheme="urn:uuid:41a5887f-8865-4c09-adf7-e362475b143a"
                        classifiedObject="urn:uuid:a1c7a9ac-83aa-4eaf-b5e3-d355b57a5016" nodeRepresentation="60591-5">
                    <rim:Slot name="codingScheme">
                            <rim:ValueList>
                                <rim:Value>2.16.840.1.113883.6.1</rim:Value>
                            </rim:ValueList>
                        </rim:Slot>
                        <rim:Name>
                            <rim:LocalizedString xml:lang="en" charset="UTF-8" value="Patient Summary"/>
                        </rim:Name>
                    </rim:Classification>
                    <!-- Type Code - Patient Summary (60591-5) -->
                    <rim:Classification
                        id="urn:uuid:c7cffb04-3537-4e8b-963d-f2f639c734de"
                        lid="urn:uuid:c7cffb04-3537-4e8b-963d-f2f639c734de" objectType="urn:oasis:names:tc:ebxml-
regrep:ObjectType:RegistryObject:Classification" classificationScheme="urn:uuid:f0306f51-975f-434e-a61c-c59651d33983"
                        classifiedObject="urn:uuid:a1c7a9ac-83aa-4eaf-b5e3-d355b57a5016" nodeRepresentation="60591-5">
                        <rim:Slot name="codingScheme">
                            <rim:ValueList>
                                <rim:Value>2.16.840.1.113883.6.1</rim:Value>
                            </rim:ValueList>
                        </rim:Slot>
                        <rim:Name>
                            <rim:LocalizedString xml:lang="en" charset="UTF-8" value="Patient Summary"/>
                        </rim:Name>
                    </rim:Classification>
                    <!-- Format Code -->
                    <rim:Classification
                        id="urn:uuid:ca064887-589c-408a-be6f-b7844f473ee6" lid="urn:uuid:ca064887-589c-408a-be6f-
                        b7844f473ee6" objectType="urn:oasis:names:tc:ebxml-
                                    regrep:ObjectType:RegistryObject:Classification"
                        classificationScheme="urn:uuid:a09d5840-386c-46f2-b5ad-9c3699a4309d"
                        classifiedObject="urn:uuid:a1c7a9ac-83aa-4eaf-b5e3-d355b57a5016" nodeRepresentation="urn:ihe:iti:xds-
                        sd:pdf:2008">
                        <rim:Slot name="codingScheme">
                            <rim:ValueList>
                                <rim:Value>MyHealth@EU formatCodes</rim:Value>
                            </rim:ValueList>
                        </rim:Slot>
                        <rim:Name>
                            <rim:LocalizedString xml:lang="en" charset="UTF-8"
                                value="PDF/A Coded Document"/>
                        </rim:Name>
```

```
                    </rim:Classification>
                    <!-- Patient ID -->
                     <rim:ExternalIdentifier
                          id="urn:uuid:27d19a5d-7850-4c37-9499-a42fe6fdd5c8"
                          lid="urn:uuid:27d19a5d-7850-4c37-9499-a42fe6fdd5c8" objectType="urn:oasis:names:tc:ebxml-
  regrep:ObjectType:RegistryObject:ExternalIdentifier" identificationScheme="urn:uuid:58a6f841-87b3-4a3e-92fd-a8ffeff98427"
                          value="90378912821^^^&amp;amp;1.3.6.1.4.1.21367.2005.3.7&amp;amp;ISO" registryObject="urn:uuid:a1c7a9ac-
                          83aa-4eaf-b5e3-d355b57a5016">
                          <rim:Name>
                               <rim:LocalizedString xml:lang="en-us" charset="UTF-8"
                                    value="XDSDocumentEntry.patientId"/>
                          </rim:Name>
                     </rim:ExternalIdentifier>
                    <!-- Unique ID -->
                     <rim:ExternalIdentifier
                          id="urn:uuid:81854cc8-2b26-45d6-8132-9f9c7eb2e5ae" lid="urn:uuid:81854cc8-2b26-45d6-8132-
                          9f9c7eb2e5ae" objectType="urn:oasis:names:tc:ebxml-
  regrep:ObjectType:RegistryObject:ExternalIdentifier"
                          identificationScheme="urn:uuid:2e82c1f6-a085-4c72-9da3-8640a32e42ab"
                          value="1.42.20100103225206.3.2"
                          registryObject="urn:uuid:a1c7a9ac-83aa-4eaf-b5e3-d355b57a5016">
                          <rim:Name>
                               <rim:LocalizedString xml:lang="en-us" charset="UTF-8"
                                    value="XDSDocumentEntry.uniqueId"/>
                          </rim:Name>
                      </rim:ExternalIdentifier><rimext:Document>
                           UjBsR09EbGhjZ0dTQUxNQUFBUUNBRU1tQ1p0dU1GUXhEUzhi....
                      </rimext:Document>
                 </rimext:ExtrinsicObject>
                 <rim:Association id="urn:uuid:f4618a30-a7fb-49a3-b27f-d1994b9c4e32" lid="urn:uuid:f4618a30-a7fb-49a3-b27f-
                     d1994b9c4e32" status="urn:oasis:names:tc:ebxml-regrep:StatusType:Approved"
                     associationType="urn:ihe:iti:2007:AssociationType:XFRM" sourceObject="urn:uuid:fbf2ea29-3aa3-4bc5-9187-
                     01d7b6b0f481" targetObject="urn:uuid:a1c7a9ac-83aa-4eaf-b5e3-d355b57a5016">
                      <rim:Classification
                          id="urn:uuid:8ec64c7e-8b7d-4d63-8741-c5a5890e5af3" lid="urn:uuid:8ec64c7e-8b7d-4d63-8741-
                          c5a5890e5af3" classificationScheme="urn:uuid:abd807a3-4432-4053-87b4-fd82c643d1f3"
                          classifiedObject="urn:uuid:a1c7a9ac-83aa-4eaf-b5e3-d355b57a5016"
                          objectType="urn:oasis:names:tc:ebxml-
   regrep:ObjectType:RegistryObject:Classification"
                          nodeRepresentation="epSOS pivot">
                          <rim:Slot name="codingScheme">
                               <rim:ValueList>
                                    <rim:Value>MyHealth@EU translation types</rim:Value>
                               </rim:ValueList>
                          </rim:Slot>
                          <rim:Name>
                               <rim:LocalizedString
                                 value="Translation into MyHealth@EU pivot format"/>
                          </rim:Name>
                     </rim:Classification>
                 </rim:Association>
            </rim:RegistryObjectList>
       </query:AdhocQueryResponse>
   </soapenv:Body>
</soapenv:Envelope>
```

3.1.4.2 Security Audit Considerations

The service consumer MUST write an audit trail entry according to the HP Assurance Audit Schema. The service provider MUST write an audit trail entry according to the Patient Privacy Audit Schema.
The following table defines which categories MUST be filled (R), which MAY be filled (O) and which categories MUST NOT be used (X).

MyHealth@EU Instance	Opt.	Description
Event	R	Audited event
Requesting Point of Care	R / X	HCPO that issued the original request. This category MUST be filled by the service consumer. It MUST NOT be provided by the service provider.
Human Requestor	R	HP that triggered the request

Source Gateway	R	Service consumer node address at the country of Care
Target Gateway	R	Service provider node address at the country of the patient's affiliation
Audit Source	R	Legal entity that ensures the uniqueness of the identifiers that are used to identify active participants
Patient	R	Patient
Event Target	R	Subject to the Query
Error Message	O	Only used in case that the request handling was not completed successfully

For the Event Target Category the following fields MUST be provided:

Field Name	Opt.	Value Constraints
ParticipantObjectTypeCode	R	MUST be "2" (System Object)
ParticipantObjectTypeCodeRole	R	MUST be "24" (Query)
ParticipantObjectIDTypeCode	R	MUST be "10" (Search Criteria)
ParticipantObjectID	R	MUST be string-encoded UUIDs of the returned documents

3.1.4.3 Protocol Requirements

The *MyHealth@EU Patient Service* List() request and response messages will be transmitted using synchronous Web Services Exchange, according to the requirements specified in section 4.3 of this document.
Port types and bindings MUST be used as defined in the WSDL given in section 6.4.2 of this document. Acc. to this the MyHealth@EU Patient Service List() operation's request and response data MUST be contained within the message body as follows:

MyHealth@EU Patient Service	Message Body
List request	CrossGatewayQueryRetrieve_Message (see section 6.4.2)
List response	CrossGatewayQueryRetrieveResponse_Message (see section 6.4.2)

The request message MUST be protected by the service consumer (NCP-B) according to the MyHealth@EU message security considerations as defined in section 4.3.5.2 of this document. The response message MUST be protected by the service provider (NCP-A) according to the MyHealth@EU message security considerations as defined in section 4.3.5.2 of this document.

3.1.5 MyHealth@EU Order Service

The MyHealth@EU Order Service is used to share an identified patient's ePrescriptions between the patient's country of affiliation and the country of care. Both countries are represented by their respective NCPs.



Figure 10 - Order Service Interface

The implementation of the *MyHealth@EU Order Service* is based on the following standards:

- ebRIM: OASIS/ebXML Registry Information Model v3.0 [OASIS ebRIM 3.0]

- ebRS: OASIS/ebXML Registry Services Specifications v3.0[8] [OASIS ebRS 3.0]
- XOP: XML-binary Optimized Packaging [W3C XOP]

and:

- XCA: IHE Cross-Community Access [IHE XCA]

For discovery and localisation of the Order Service instance that is responsible for providing access to the identified patient's data see section 2.1.4 of this document.

3.1.5.1 List() Operation

The *MyHealth@EU Order Service* list() operation is implemented as an IHE XCA Cross-Community Access transaction. It is fully compliant with the ebRS 3.0 standard. The *MyHealth@EU Order Service* list() operation includes the documents listed in the response meta-data, just like they would have been included in Cross-Gateway Retrieve.

3.1.5.2 Request Message

The list() request is initiated by an HP in the country of care for retrieving the available ePrescription documents of an identified patient. The respective request message builds upon the IHE XCA Cross-Community Access request message.

The <AdhocQueryRequest/> element that encapsulates the query parameters MUST be used as follows for MyHealth@EU:

Element Name	MyHealth@EU Usage Convention
ResponseOption/@returnComposedObjects	MUST be "true"
ResponseOption/@returnType	MUST be "leafClassWithRepositoryItem"
AdhocQuery	Container for holding the ebML stored query arguments. All arguments MUST be encoded as query slots (see table below).
AdhocQuery@id	MUST be "urn:uuid:f2072993-9478-41df-a603-8f016706efe8" which indicates a Fetch (which is an adaption of the findDocuments Query as defined in ITI TF-2a:3.18.1)

Only synchronous web services exchange MUST be used. The *XDS Affinity Domain Option* only applies to the national environment. Therefore it MUST NOT be used for NCP-2-NCP message exchange.

Stored query argument slots MUST be defined for the patient identifier and the document class code. The document format code and the document type code MAY be given. Other argument slots than the ones listed below MUST be ignored by the service provider and SHOULD NOT be issued by the service consumer.

Slot Name	Opt	Slot Value
$XDSDocumentEntryPatientId	R	Equals to the patient identifier that was provided by the *MyHealth@EU Identification Service* (encoded as HL7 v3 II data type)
$XDSDocumentEntryStatus	R	Only approved documents MUST be returned: 'urn:oasis:names:tc:ebxml-regrep: StatusType:Approved'
$XDSDocumentEntryClassCode	R	ePrescription LOINC code ("57833-6") coded according to specification in ITI TF-2a: 3.18.4.1.2.3.4 Coding of Code/Code-Scheme. As classification scheme 2.16.840.1.113883.6.1 MUST be used:

[8] The integration of ebRS and MTOM as used by MyHealth@EU is not compatible with the current version of OASIS ebRS.

$XDSDocumentEntryTypeCode	O	ePrescription LOINC code ("57833-6") coded according to specification in ITI TF-2a: 3.18.4.1.2.3.4 Coding of Code/Code-Scheme. As classification scheme 2.16.840.1.113883.6.1 MUST be used:
$XDSDocumentEntryFormatCode	O	Format qualifier as defined in [CDA templates]; see table below for details on applying these codes to the retrieval of a patient's available ePrescriptions. Only encodings of ePrescription documents that comply to the requested format code will be returend by the service provider. If this stored query slot is omitted, the service provider MUST respond with all available encodings.

For the document format only the format codes defined in [CDA templates] and listed in the following table MUST be used.

Document Format	Format Code	Document content
MyHealth@EU pivot coded ePrescription	urn:epSOS:ep:pre:2010	HL7 CDA document acc. [CDA templates]. The patient's country of affiliation MUST be able to provide the patient's available ePrescriptions in this format.
PDF/A source coded document	urn:ihe:iti:xds-sd:pdf:2008	CDA-enveloped PDF/A encoding of the original document without any semantic transformation. The patient's country of affiliation MUST be able to provide the patient's available ePrescriptions in this format.

3.1.5.2.1 Example Request Message

The following excerpt from a MyHealth@EU Order Service list() request message shows an IHE XCA based Cross-Community Access request that contains argument slots for retrieving the available ePrescriptions (LOINC code 57833-6) of an identified patient (patient identifier 90378912821). In this example the service consumer does not specify the requested encoding. Therefore the service provider MUST deliver both encodings (MyHealth@EU pivot and PDF/A) for all available ePrescriptions.

```
<soapenv:Envelope xmlns:soapenv=" http://www.w3.org/2003/05/soap-envelope" ... >
<soapenv:Header> ... </soapenv:Header>
<soapenv:Body>
<query:AdhocQueryRequest>
    <query:ResponseOption returnComposedObjects="true" returnType="LeafClassWithRepositoryItem"/>
    <rim:AdhocQuery id="urn:uuid:14d4debf-8f97-4251-9a74-a90016b0af0d">
        <rim:Slot name="$XDSDocumentEntryPatientId">
         <rim:ValueList>
                <rim:Value>
                    '90378912821^^^&amp;1.3.6.1.4.1.21367.2005.3.7&amp;ISO'
                </rim:Value>
         </rim:ValueList>
        </rim:Slot>
        <rim:Slot name="$XDSDocumentEntryStatus">
        <rim:ValueList>
                <rim:Value>
                    ('urn:oasis:names:tc:ebxml-regrep:StatusType:Approved')
                </rim:Value>
        </rim:ValueList>
        </rim:Slot>
        <rim:Slot name="$XDSDocumentEntryClassCode">
        <rim:ValueList>
                <rim:Value>('57833-6^^2.16.840.1.113883.6.1')
        </rim:Value>
        </rim:ValueList>
        </rim:Slot>
</rim:AdhocQuery>
    </query:AdhocQueryRequest>
```

```
</soapenv:Body>
</soapenv:Envelope>
```

3.1.5.2.2 Expected Actions

The *MyHealth@EU Order Service* provider shall respond to a ListRequest message with the ListResponse message containing

- the identified patient's available ePrescriptions together with a status notification (full success scenario, see section 3.4.1.4) or
- an error message (no ePrescriptions provided, see section 3.4.1.5).

The *MyHealth@EU Order Service* provider MUST verify that the requesting service user has sufficient rights to access the available ePrescriptions of the identified patient. In case of an error that relates to the transmission of the request or the processing of the MyHealth@EU security token, the *MyHealth@EU Order Service* provider MUST respond with a fault message according to section 4.4 of this document.

3.1.5.2.3 Response Message (Full Success Scenario)

Depending on the requested format code the MyHealth@EU list() response contains the MyHealth@EU pivot encoded ePrescription documents, the PDF/A source coded ePrescription documents of the identified patient or both sets of documents. If both encodings are provided, a 1:1 association between any source coded PDF document and its derived MyHealth@EU pivot CDA coded document MUST be given.
The respective message builds upon to the IHE XCA Cross-Gateway Fetch response and Cross-Gateway Fetch Response messages, by creating a new combined QueryRetrieve-alike message. The fields defined for the *MyHealth@EU Order Service* ListResponse message MUST be used as follows:

Element Name	MyHealth@EU Usage Convention
Query:AdhocQueryResponse	Response message acc to IHE XCA Cross-Gateway Access response message [IHE XCA]
@status	For the full success scenario the response status MUST be set to "urn:oasis:names:tc:ebxml-regrep:ResponseStatusType:Success" or "urn:ihe:iti:2007:ResponseStatusType:PartialSuccess" (for details see table below)
rs:RegistryErrorList	In case that a warning is given by the service provider, this element holds the respective warning codes and messages. It must be used acc. to section 4.1.13 of [IHE ITI TF-3]
rim:RegistryObjectList	This element MUST be provided for the full success scenario. It MUST
rim:ExtrinsicObject	For each instance of a ePrescription document a <rim:ExtrinsicObject/> element MUST be provided. Each <rim:ExtrinsicObject/> element is described and classified by metadata acc. to Table 4 below.
rimext:Document	This element MUST appear as the last element child of an <rim:ExtrinsicObject/> element. It may appear zero or one times. This element contains the base 64 encoded content of the document. The document contents are associated with the DocumentEntry (ExtrinsicObject) metadata by the fact that it is nested inside it within the XML. The base64 encoded document content MAY be encrypted. How encryption is applied and how the encryption key is negotiated should be subject to an additional specification on advanced security safeguards.

Each provided ePrescription document and each of its encodings (MyHealth@EU pivot and/or source coded PDF) MUST be further classified by metadata. The following table

lists the usage conventions that have to be followed for the *MyHealth@EU Order Service* response message. If not stated otherwise the classification schemes as defined in section 4.3.1.2 of [IHE ITI TF-3] MUST be used. If no restrictions on metadata values are given, the metadata elements MUST be used as per [IHE XCA].

Metadata (ebRIM names)	Binding	Opt.	MyHealth@EU usage convention
status	Attribute	R	MUST be "urn:oasis:names:tc:ebxml-regrep:StatusType:Approved"
mimeType	Attribute	R	MUST be "text/xml" for both MyHealth@EU pivot CDA and CDA-wrapped PDF
Name	Main	R	MAY be empty. MUST be ignored by the service consumer.
Description	Main	R	Must contain Product Name/Generic Substance Name, Dosage Form and Strength.
VersionInfo	Main	R	MUST be "1"
creationTime	rim:Slot	O	MAY be omitted by the service provider and MAY be ignored by the service consumer. If given, the value MUST be encoded as HL7 v2 Date Time "YYYY[MM[DD[hh[mm[ss]]]]]"
hash	rim:Slot	O	SHOULD be omitted by the service provider and MUST NOT be processed by the service
languageCode	rim:Slot	O	SHOULD be omitted by the service provider and MUST NOT be processed by the service consumer.
repositoryUniqueId	rim:Slot	O	MAY be omitted by the service provider and MAY be processed by the service consumer in an IHE-compatible NI-scenario.
serviceStartTime serviceEndTime	rim:Slot	O	SHOULD be omitted by the service provider and MUST NOT be processed by the service consumer.
size	rim:Slot	O	SHOULD be omitted by the service provider and MUST NOT be processed by the service consumer.
sourcePatientId	rim:Slot	R	MUST contain the same value as XDSDocumentEntry.PatientId (see below).
sourcePatientInfo	rim:Slot	X	MUST NOT be used. Future versions of MyHealth@EU MAY define different protection levels for metadata and documents. Therefore all metadata elements that might carry medical or social information MUST be omitted.
classCode	Classification	R	Patient summary LOINC code ("57833-6"). As classification scheme "urn:oid:2.16.840.1.113883.6.1" MUST be used

eventCodeList	Classification	O	ClassficationScheme: urn:uuid:2c6b8cb7-8b2a-4051-b291-b1ae6a575ef4 NodeRepresentation="ATC-code code" Valuelist/Value=2.16.840.1.113883.6.73 Name="ATC-code Display Name" MAY be omitted by the service provider (only in the case that the product does not have an ATC-code) and MAY be ignored by the service consumer.
eventCodeList	Classification	R	ClassficationScheme: urn:uuid:2c6b8cb7-8b2a-4051-b291-b1ae6a575ef4 • code: urn:ihe:iti:xdw:2011:eventCode:open codingScheme: 1.3.6.1.4.1.19376.1.2.3 • code: urn:ihe:iti:xdw:2011:eventCode:closed codingScheme: 1.3.6.1.4.1.19376.1.2.3 If the ePrescription is not dispensable the eventCode "Closed" MUST be used. Otherwise the eventCode "Open" MUST be used. MAY be ignored by the service consumer. Therefore measures to disallow the prescription of Non-dispensable products must be in place if the list contains these non-dispensable products.
author	Classification	O	classificationScheme: urn:uuid:93606bcf-9494-43ec-9b4e-a7748d1a838d Slot name="authorPerson" MAY be used by the service provider and MAY be ignored by the service consumer. But it is useful to identify the right prescription on the service consumer side.
confidentialityCode	Classification	R	MUST be provided for XCA compatibility but MAY be ignored by the service consumer. Value
formatCode	Classification	R	MUST be "urn:epSOS:ep:pre:2010" for MyHealth@EU pivot CDA and "urn:ihe:iti:xds-sd:pdf:2008" for MyHealth@EU source coded PDF (see [CDA templates]).
healthcareFacilityTypeCode	Classification	R	MUST be provided for XCA compatibility and correct addressing. Value MUST be set to ISO 3166-1 alpha-2 country code of the addressed PN.
practiceSettingCode	Classification	R	MUST be provided for XCA compatibility. Value MUST be set to "Not Used" in order to protect private patient information.
XDSDocumentEntry.uniqueId	ExternalIdentifier	R	MUST hold the OID of the document. The document unique id value MUST be the same as the value of the document's <ClinicalDocument/id> CDA header element.
XDSDocumentEntry.patientId	ExternalIdentifier	R	MUST hold the patient identifier. The service consumer MUST verify that this id matches the patient Id that was discovered by the MyHealth@EU Identification Service.

Table 4: MyHealth@EU ePrescription Metadata

Other metadata than the ones listed above MUST NOT be provided by the service provider.

Multiple ePrescriptions (with up to two encodings) MAY be available per patient. An ebRIM association MUST be used for declaring the MyHealth@EU pivot coded document as a transformation of the source coded document. As classification scheme urn:uuid:abd807a3-4432-4053-87b4-fd82c643d1f3 MUST be used per IHE-XCA. "MyHealth@EU pivot" is defined as the only code value for this MyHealth@EU valid transformation.

```
        <rim:Association id="id of the association" associationType="urn:ihe:iti:2007:AssociationType:XFRM"
        sourceObject="UUID of the source coded document" targetObject="UUID of the MyHealth@EU pivot document"
        objectType="urn:oasis:names:tc:ebxml-regrep:ObjectType:RegistryObject:Classification">
<rim:Classification id="id of the classification" classificationScheme="urn:uuid:abd807a3-4432-4053-87b4-fd82c643d1f3"
classifiedObject="id of the association"
objectType="urn:oasis:names:tc:ebxml-regrep:ObjectType:RegistryObject:Classification"
nodeRepresentation="epSOS pivot">
 <rim:Name>
  <rim:LocalizedString value="Translation into MyHealth@EU pivot format"/>
 </rim:Name>
 <rim:Slot name="codingScheme">
  <rim:ValueList>
  <rim:Value>MyHealth@EU translation types</rim:Value>
  </rim:ValueList>
 </rim:Slot>
 <rim:name>
  <rim:LocalizedString value="Translation into MyHealth@EU pivot format" />
 </rim:Name>
 </rim:Classification>
</rim:Association>
```

If a warning is to be transmitted to the HP (see section 3.1.4) the ebXML Registry Error mechanism MUST be used with a syntax as defined in section 3.43.5 of [IHE ITI TF-2]. The table from Appendix 6.4 lists the MyHealth@EU defined warning codes.

3.1.5.2.4 Response Message (No ePrescriptions Provided)

If the MyHealth@EU Order Service provider is unable to respond with the patient's ePrescription data in the requested encoding it MUST respond with a ListResponse message that only contains a <RetrieveDocumentSetResponse/RegistryResponse> element.

For a full list of error messages defined for IHE X* see table 4.1-11 in [IHE ITI TF-3]. The table from Appendix 6.4 lists the additional, MyHealth@EU-specific response status types and error/warning/info codes to be used within the <RegistryErrorList> element.

Example Response Message

In this section three possible response messages to the previously sketched request message are shown.

The first example response message covers the case where a single ePrescription is discovered and provided as both MyHealth@EU pivot and source PDF encoding. As the MyHealth@EU Order Service list() response message is very similar to the MyHealth@EU Patient Service list() response message (see section 3.3.1.6 for an example) only an excerpt is shown.

```
<soapenc:Envelope>
<soapenv:Header>....</soapenv:Header>
 <soapenv:Body>
    <query:AdhocQueryResponse
            status="urn:oasis:names:tc:ebxml-regrep:ResponseStatusType:Success">
                                        <rim:RegistryObjectList>
        <!—MyHealth@EU source coded CDA wrapped PDF ePrescription document -->
        <rimext:ExtrinsicObject id="urn:uuid:cf614a65-d214-4b0d-b4b8-a0be3888f847" lid="urn:uuid:cf614a65-d214-4b0d-
            b4b8-a0be3888f847" objectType="urn:uuid:7edca82f-054d-47f2-a032-9b2a5b5186c1"
            status="urn:oasis:names:tc:ebxml-regrep:StatusType:Approved" mimeType="text/xml">
            <!-- metadata missing here (see Patient Service example) --><rimext:Document
                >UjBsR09EbGhjZ0dTQUxNQUFBUUNBRU1tQ1p0dU1GUXhEUzhi</rimext:Document>
                                        </rimext:ExtrinsicObject>
        <!—MyHealth@EU source coded CDA Pivot ePrescription document -->
```

```
            <rimext:ExtrinsicObject id="urn:uuid:eec764cf-9fe5-4101-8e86-33a13fb06e4a" lid="urn:uuid:eec764cf-9fe5-4101-
                8e86-33a13fb06e4a" objectType="urn:uuid:7edca82f-054d-47f2-a032-9b2a5b5186c1"
                status="urn:oasis:names:tc:ebxml-regrep:StatusType:Approved" mimeType="text/xml">

                <!-- metadata missing here (see Patient Service example) --><rimext:Document
                    >UjBsR09EbGhjZ0dTQUxNQUFBUUNBRU1tQ1p0dU1GUXhEUzhi</rimext:Document>
                                                    </rimext:ExtrinsicObject>
            <rim:Association id="urn:uuid:b4fc4809-0096-4b76-a7b1-3135ac5e5614" lid="urn:uuid:b4fc4809-0096-4b76-a7b1-
                3135ac5e5614" associationType="urn:oasis:names:tc:ebxml-regrep:AssociationType:XFRM"
                sourceObject="urn:uuid:eec764cf-9fe5-4101-8e86-33a13fb06e4a" targetObject="urn:uuid:cf614a65-d214-4b0d-
                b4b8-a0be3888f847" objectType="urn:oasis:names:tc:ebxml-regrep:ObjectType:RegistryObject:Classification">
                <rim:Classification   id="urn:uuid:969d2f2b-5f5a-4c24-af0a-d07d16ddaeb9"
                    classificationScheme="urn:uuid:abd807a3-4432-4053-87b4-fd82c643d1f3"
                    classifiedObject="urn:uuid:b4fc4809-0096-4b76-a7b1-3135ac5e5614"
                    objectType="urn:oasis:names:tc:ebxml-
regrep:ObjectType:RegistryObject:Classification" nodeRepresentation="epSOS pivot">
                    <rim:Slot name="codingScheme">
                        <rim:ValueList>
                            <rim:Value>MyHealth@EU translation types</rim:Value>
                        </rim:ValueList>
                    </rim:Slot>
                    <rim:Name>
                        <rim:LocalizedString value="Translation into MyHealth@EU pivot format"/>
                    </rim:Name>
                </rim:Classification>
            </rim:Association>
        </rim:RegistryObjectList>
      </query:AdhocQueryResponse>
  </soapenv:Body>
 </soapenc:Envelope>
```

The second example response message shows the case where two ePrescriptions are discovered. The first one is provided as MyHealth@EU pivot and source PDF encoding. For the second one country-A is not able to transform an ePrescription to the MyHealth@EU pivot format. Only the PDF encoding is provided and an information given, that MyHealth@EU pivot transcoding failed for this ePrescription[9].

```
        <soapenc:Envelope>
        <soapenv:Header>....</soapenv:Header>
        <soapenv:Body>
            <query:AdhocQueryResponse
                    status="urn:oasis:names:tc:ebxml-regrep:ResponseStatusType:PartialSuccess">
              <rs:RegistryErrorList>
                <rs:RegistryError
                    severity="urn:oasis:names:tc:ebxml-regrep:ErrorSeverityType:Error" errorCode="2104"
                    codeContext="Rendering incomplete"/ location="urn:uuid:a1c7a9ac-83aa-4eaf-b5e3-d355b57a5016">
                                                            </rs:RegistryErrorList>
                                                          <rim:RegistryObjectList>
                <!-- ePrecription 1: MyHealth@EU source coded CDA wrapped PDF -->
                <rimext:ExtrinsicObject id="urn:uuid:cf614a65-d214-4b0d-b4b8-a0be3888f847" lid="urn:uuid:cf614a65-d214-4b0d-
                    b4b8-a0be3888f847" objectType="urn:uuid:7edca82f-054d-47f2-a032-9b2a5b5186c1"
                    status="urn:oasis:names:tc:ebxml-regrep:StatusType:Approved" mimeType="text/xml">
                    <!-- metadata and contents missing here (see Patient Service example) -->
                </rimext:ExtrinsicObject>
                <!-- ePrecription 1: MyHealth@EU source coded CDA Pivot document -->
                <rimext:ExtrinsicObject id="urn:uuid:eec764cf-9fe5-4101-8e86-33a13fb06e4a" lid="urn:uuid:eec764cf-9fe5-4101-
                    8e86-33a13fb06e4a" objectType="urn:uuid:7edca82f-054d-47f2-a032-9b2a5b5186c1"
                    status="urn:oasis:names:tc:ebxml-regrep:StatusType:Approved" mimeType="text/xml">
                    <!-- metadata and content missing here (see Patient Service example) -->
                </rimext:ExtrinsicObject>
                <!-- ePrecription 2: MyHealth@EU source coded CDA wrapped PDF document -->
                <rimext:ExtrinsicObject id="urn:uuid:a1c7a9ac-83aa-4eaf-b5e3-d355b57a5016" lid="urn:uuid:a1c7a9ac-83aa-4eaf-
                    b5e3-d355b57a5016" objectType="urn:uuid:7edca82f-054d-47f2-a032-9b2a5b5186c1"
                    status="urn:oasis:names:tc:ebxml-regrep:StatusType:Approved" mimeType="text/xml">
                    <!-- metadata missing here (see Patient Service example) --><rimext:Document
                            >UjBsR09EbGhjZ0dTQUxNQUFBUUNBRU1tQ1p0dU1GUXhEUzhi</rimext:Document>
                </rimext:ExtrinsicObject>
                <!-- Association for ePrescription 1; for ePrescription 2 no association is defined because only one encoding is
                        provided -->
```

[9] It's up to country A to decide on how to act in case of a failed MyHealth@EU pivot translation. This example covers the case where country A transmits the source coded document only. This may e.g. make sense in cases where both country A and B share a common language.

```
                <rim:Association id="urn:uuid:b4fc4809-0096-4b76-a7b1-3135ac5e5614" lid="urn:uuid:b4fc4809-0096-4b76-a7b1-
                    3135ac5e5614" associationType="urn:oasis:names:tc:ebxml-regrep:AssociationType:XFRM"
                    sourceObject="urn:uuid:eec764cf-9fe5-4101-8e86-33a13fb06e4a" targetObject="urn:uuid:cf614a65-d214-4b0d-
                    b4b8-a0be3888f847" objectType="urn:oasis:names:tc:ebxml-regrep:ObjectType:RegistryObject:Classification">
                    <rim:Classification   id="urn:uuid:969d2f2b-5f5a-4c24-af0a-d07d16ddaeb9"
                            classificationScheme="urn:uuid:abd807a3-4432-4053-87b4-fd82c643d1f3"
                            classifiedObject="urn:uuid:b4fc4809-0096-4b76-a7b1-3135ac5e5614"
                            objectType="urn:oasis:names:tc:ebxml-
regrep:ObjectType:RegistryObject:Classification" nodeRepresentation="epSOS pivot">
                            <rim:Slot name="codingScheme">
                                <rim:ValueList>
                                    <rim:Value>MyHealth@EU translation types</rim:Value>
                                </rim:ValueList>
                            </rim:Slot>
                            <rim:Name>
                                <rim:LocalizedString value="Translation into MyHealth@EU pivot format"/>
                            </rim:Name>
                        </rim:Classification>
                    </rim:Association>
                </rim:RegistryObjectList>
            </query:AdhocQueryResponse>
        </soapenv:Body>
</soapenv:Envelope>
```

3.1.5.3 Security Audit Considerations

The service consumer MUST write an audit trail entry according to the HP Assurance Audit Schema. The service provider MUST write an audit trail entry according to the *Patient Privacy Audit Schema*.
The following table defines which categories MUST be filled (R), which MAY be filled (O) and which categories MUST NOT be used (X).

MyHealth@EU Instance	Opt.	Description
Event	R	Audited event
Requesting Point of Care	R / X	HCPO that issued the original request. This category MUST be filled by the service consumer. It MUST NOT be provided by the service provider.
Human Requestor	R	HP that triggered the request
Source Gateway	R	Service consumer node address at the country of Care
Target Gateway	R	Service provider node address at the country of the patient's affiliation
Audit Source	R	Legal entity that ensures the uniqueness of the identifiers that are used to identify active participants
Patient	R	Patient
Event Target	R	Subject to the Query
Error Message	O	Only used in case that the request handling was not completed successfully

For the Event Target Category the following fields MUST be provided:

Field Name	Opt.	Value Constraints
ParticipantObjectTypeCode	R	MUST be "2" (System Object)
ParticipantObjectTypeCodeRole	R	MUST be "24" (Query)
ParticipantObjectIDTypeCode	R	MUST be "10" (Search Criteria)
ParticipantObjectID	R	MUST be string-encoded UUIDs of the returned documents

3.1.5.4 Protocol Requirements

The MyHealth@EU Order Service List() request and response messages will be transmitted using synchronous Web Services Exchange, according to the requirements specified in section 4.3 of this document.

Port types and bindings MUST be used as defined in the WSDL given in section 6.4.2 of this document. Acc. to this the *MyHealth@EU Order Service* List() operation's request and response data MUST be contained within the message body as follows:

MyHealth@EU Order Service	**Message Body**
List request	CrossGatewayQueryRetrieve_Message (see section 6.4.2)
List response	CrossGatewayQueryRetrieveResponse_Message (see section 6.4.2)

The request message MUST be protected by the service consumer (NCP-B) according to the MyHealth@EU message security considerations as defined in section 4.3.5.2 of this document. The response message MUST be protected by the service provider (NCP-A) according to the MyHealth@EU message security considerations as defined in section 4.3.5.2 of this document.

3.1.6 MyHealth@EU Dispensation Service

The *MyHealth@EU Dispensation Service* is used to share an identified patient's eDispensation data between the patient's country of affiliation and the country of care. Both countries are represented by their respective NCPs.



Figure 11: Dispensation Service Interface

The implementation of the *MyHealth@EU Dispensation Service* is based on the following standards:

- ebRIM: OASIS/ebXML Registry Information Model v3.0 [OASIS ebRIM 3.0]
- ebRS: OASIS/ebXML Registry Services Specifications v3.0 [OASIS ebRS 3.0]
- XOP: XML-binary Optimized Packaging [W3C XOP]

and is compliant with the IHE profiles:

- XDR: IHE Cross-Enterprise Reliable Exchange [IHE ITI TF-1] [IHE ITI TF-2]

For discovery and localisation of the *MyHealth@EU Dispensation Service* instance that is responsible for providing access to the identified patient's data see section 2.1.4 of this document.

3.1.6.1 Initialize() Operation

The *MyHealth@EU Dispensation Service* initialize() operation is implemented by the *IHE Provide And Register DocumentSet* transaction (ITI-41) as described in [IHE XDR].

3.1.6.2 Request Message

The initialize() request is initiated by an HP in the country of care for handing over dispensation notifications to the patient's country of affiliation. Each dispensation notification consists of an MyHealth@EU pivot coded eDispensation document acc. to [CDA templates] and the source coded document that encodes the same information

without semantic mapping. An initialize() request MAY contain multiple MyHealth@EU coded and source coded documents.
The *MyHealth@EU Dispensation Service* InitializeRequest message is a specialisation of the *IHE Provide And Register DocumentSet* transaction (ITI-41) request message as profiled in [IHE XDR].
The fields defined for the *ProvideAndRegisterDocumentSetRequest* message MUST be used as follows:

Element Name	MyHealth@EU Usage Convention
lcm:SubmitObjectsRequest	Container that can be used to provide the metadata for the transmitted documents, the submission set and the associations between documents (see below).
rim:RegistryObjectList	Container that contains (pointers to) all eDispensation documents
rim:ExtrinsicObject	For each eDispensation document a single extrinsic object MUST be defined. There MUST be a 1:1 id-correspondence between <rim:ExtrinsicObject> elements and <ihe:Document> elements. For a list of further metadata to be provided with an eDispensation document see the table below.
rimext:Document	base64 encoded data for the eDispensation documents being submitted to the service provider. The <rimext:Document/> element also includes the document id attribute (rimext:Document/@id) of type xsd:anyURI to match the document ExtrinsicObject id in the metadata and providing the necessary linkage. The base64 encoded document content MAY be encrypted. How encryption is applied and how the encryption key is negotiated should be subject to an additional specification on advanced security safeguards.
rim:Association	For each pair of MyHealth@EU coded and source coded documents an ebRIM association MUST be defined (see below for details on the encoding).

The service consumer SHOULD embrace the provided documents as a single IHE XDS submission set acc. to [IHE ITI TF-2a]. The service consumer SHOULD ignore this grouping and MUST ignore all associations between documents and submission sets. The service consumer MUST NOT process any metadata assigned to the submission set, it MUST solely rely on the document metadata and contents.
For each eDispensation document (either MyHealth@EU coded or source coded) the following set of metadata MUST be provided:

Slot Name	Binding	Slot Value
id	Attribute	Identifer of the document. This identifier MUST be the same for <rim:ExtrinsicObject/@id> and <ihe:Document/@id>.
mimeType	Attribute	MUST be "text/xml"
objectType	Attribute	MUST be set acc. to section 4.3.1.2 of [IHE IT TF 3]
status	Attribute	MUST be "urn:oasis:names:tc:ebxml-regrep:StatusType:Approved"

creationTime	rim:Slot	MUST be given for XDR compatibility. MAY be ignored by the service provider.
languageCode	rim:Slot	MUST be given for XDR compatibility. MAY be ignored by the service provider.
sourcePatientID	rim:Slot	MUST be of the same value as $XDSDocumentEntryPatientId (see below)
healthcareFacilityTypeCode	classification	MUST be provided for XCA compatibility and correct addressing. Value MUST be set to ISO 3166-1 alpha-2 country code of the addressed PN.
practiceSettingCode	classification	MUST be provided for XDR compatibility but MAY be ignored by the service consumer. Value
confidentialityCode	classification	MUST be provided for XDR compatibility but MAY be ignored by the service consumer. Value SHOULD be set to "N", as long as the Minimal Metadata Profile is not published.
XDSDocumentClassCode	classification	eDispensation LOINC code ("60593-1")[10] coded according to specification in ITI TF-2a: 3.18.4.1.2.3.4 Coding of Code/Code-Scheme. As classification scheme 2.16.840.1.113883.6.1 MUST be used.
XDSDocumentFormatCode	classification	Format qualifier as defined in [CDA templates];
$XDSDocumentEntryPatientId	External identifier	Equals to the patient identifier that was provided by the *MyHealth@EU Identification Service* (encoded as HL7 v3 II data type)
$XDSDocumentUniqueId	External identifier	MUST refer to the OID of the CDA document that is included within the <ihe:Document> element.

Other metadata than the ones listed above SHOULD NOT be provided by the service provider[11]. If given they MUST be ignored by the service consumer.
An ebRIM association MUST be used for declaring the MyHealth@EU pivot coded eDispensation document as a transformation of the source coded eDispensation document. As classification scheme urn:uuid:abd807a3-4432-4053-87b4-fd82c643d1f3 MUST be used per IHE-XCA. Currently "MyHealth@EU pivot" is defined as the only valid transformation:

```
<rim:Association id="id of the association" associationType="urn:ihe:iti:2007:AssociationType:XFRM" sourceObject="UUID
of the source coded document" targetObject="UUID of the MyHealth@EU pivot document"
        objectType="urn:oasis:names:tc:ebxml-regrep:ObjectType:RegistryObject:Classification">
        <rim:Classification id="id of the classification"
        classificationScheme="urn:uuid:abd807a3-4432-4053-87b4-fd82c643d1f3" classifiedObject="id of the association"
        objectType="urn:oasis:names:tc:ebxml-regrep:ObjectType:RegistryObject:Classification" nodeRepresentation="epSOS
        pivot">
            <rim:Slot name="codingScheme">
                <rim:ValueList>
                    <rim:Value>MyHealth@EU translation types</rim:Value>
                </rim:ValueList>
            </rim:Slot>
            <rim:Name>
```

[10] This code is a dummy that will be used for the intial NCP integration tests until a eDispensation LOINC code is approved.

[11] Document linkage information (e.g. a reference to the ePrescription that is affected by the dispensation) MUST be included with the document (see [CDA templates]) and MUST NOT be part of the message metadata.

```
                <rim:LocalizedString value="Translation into MyHealth@EU pivot format"/>
            </rim:Name>
        </rim:Classification>
</rim:Association>
```

3.1.6.2.1 Expected Actions

The *MyHealth@EU Dispensation Service* provider shall respond to an InitializeRequest message with the InitializeResponse message containing a success indicator.
The *MyHealth@EU Dispensation Service* provider MUST verify that the requesting service user has sufficient rights to submit an eDispensation for the identified patient. It MUST verify that the eDispensation matches with an ePrescription that was issued for the identified patient.
In case of an error that relates to the transmission of the request or the processing of the MyHealth@EU security token, the *MyHealth@EU Dispensation Service* provider MUST respond with a fault message according to section 4.4 of this document.

3.1.6.2.2 Response Message (Full Success Scenario)

If the *MyHealth@EU Dispensation Service* provider is able to decode the received message and to properly process all transmitted eDispensations it responds with an *ebXML Registry Response* with its status set to "urn:oasis:names:tc:ebxml-regrep:ResponseStatusType:Success"
If the service provider wants to respond with further information on the processing of the transmitted data or with a non-critical warning it SHOULD include an additional <RegistryErrorList> element. The severity MUST be set to "urn:oasis:names:tc:ebxml-regrep:ErrorSeverityType:Warning":

```
<rs:RegistryResponse status="urn:oasis:names:tc:ebxml-regrep:ResponseStatusType:Success">
    <rs:RegistryErrorList>
            <rsRegistryError severity="urn:oasis:names:tc:ebxml-regrep:ErrorSeverityType:Warning"
              errorCode="...." codeContext="Processing deferred" location="" />
    </rs:RegisryErrorList>
</rs:RegistryResponse>
```

The warning messages and codes are defined in the table Appendix 6.4.

3.1.6.2.3 Response Message (Failure or Partial Failure Scenario)

If the *MyHealth@EU Dispensation Service* provider is able to decode the received message but the processing of one or more dispensations failed, it responds with an *ebXML Registry Response* that contains a respective status indicator (see below). The response MUST contain a RegistryErrorList element that indicates the failure condition.
If none of the eDispensations was processed succesfully, the response status MUST be set to "urn:oasis:names:tc:ebxml-regrep:ResponseStatusType:Failure". If at least one eDispensation was processed successfully, the response status MUST be set to "urn:ihe:iti:2007:ResponseStatusType:PartialSuccess".
A failure location MUST be provided if the error does not apply to all provided eDispensation documents. It MUST NOT be given if the error applies to all provided documents.
The severity of each registry error message MUST be set to "urn:oasis:names:tc:ebxml-regrep:ErrorSeverityType:Error". Multiple registry error messages MAY be included within a single <rs:RegistryErrorList> element.
MyHealth@EU error messages are defined into the from Appendix 6.4 in addition of the XDS-b error messages defined in Table 4.1-11 of [IHE ITI TF-3]

3.1.6.2.4 Example Response Message

The following example shows a possible positive resonse to the request given in section 3.5.1.2:

```
<?xml version="1.0" encoding="ISO-8859-1" standalone="yes"?>
<env:Envelope xmlns:env="http://www.w3.org/2003/05/soap-envelope">
 <env:Header>
      <Action xmlns="http://www.w3.org/2005/08/addressing">urn:ihe:iti:2007:ProvideAndRegisterDocumentSet-
bResponse</Action>
      <MessageID xmlns="http://www.w3.org/2005/08/addressing">uuid:98f4bf0c-f21a-48bc-8518-
958c9d9dc4c11</MessageID>
      <RelatesTo xmlns="http://www.w3.org/2005/08/addressing">uuid:98f4bf0c-f21a-48bc-8518-
958c9d9dc4c1</RelatesTo>
   </env:Header>
   <env:Body>
      <RegistryResponse status="urn:oasis:names:tc:ebxml-regrep:ResponseStatusType:Success"
xmlns="urn:oasis:names:tc:ebxml-regrep:xsd:rs:3.0"/>
</env:Body>
</env:Envelope>
```

The following example shows a possible negative response to the request given in section 3.5.1.2:

```
<?xml version="1.0" encoding="UTF-8"?>
<soapenv:Envelope xmlns=...>
<soapenv:Header>...</soapenv:header>
<soapenv:Body>
    <rs:RegistryResponse
                status="urn:oasis:names:tc:ebxml-regrep:ResponseStatusType:Failure">
      <rs:RegistryErrorList>
      <rs:RegistryError
                severity="urn:oasis:names:tc:ebxml-regrep:ErrorSeverityType:Error"
                errorCode="...." codeContext="No Match" location="1.42.20100103225206.3.3" />
       </rs:RegistryErrorList>
    </rs:RegistryResponse>
    </soapenv:Body>
    </soapenv:Envelope>
```

3.1.6.3 Security Audit Considerations

The service consumer MUST write an audit trail entry according to the *HP Assurance Audit Schema*. The service provider MUST write an audit trail entry according to the *Patient Privacy Audit Schema*.

The following table defines which categories MUST be filled (R), which MAY be filled (O) and which categories MUST NOT be used (X).

MyHealth@EU Instance	Opt.	Description
Event	R	Audited event
Requesting Point of Care	R / X	HCPO that issued the original request. This category MUST be filled by the service consumer. It MUST NOT be provided by the service provider.
Human Requestor	R	HP that triggered the request
Source Gateway	R	Service consumer node address at the country of Care
Target Gateway	R	Service provider node address at the country of the patient's affiliation
Audit Source	R	Legal entity that ensures the uniqueness of the identifiers that are used to identify active participants
Patient	R	Patient
Event Target	R	References to the provided dispensation documents (see below)

Error Message	O	Only used in case that the request handling was not completed successfully

For the Event Target Category the following fields MUST be provided:

Field Name	Opt.	Value Constraints
ParticipantObjectTypeCode	R	MUST be "2" (System Object)
ParticipantObjectTypeCodeRole	R	MUST be "4" (Resource)
ParticipantObjectIDTypeCode	R	MUST be "12" (URI)
ParticipantObjectID	R	MUST be string-encoded UUIDs of the provided documents

3.1.6.4 Discard() Operation

The MyHealth@EU Dispensation Service discard() operation can be used to deprecate a previously transmitted eDispensation. The MyHealth@EU Dispensation Service initialize() operation is implemented by the IHE Provide And Register DocumentSet transaction (ITI-41) as described in [IHE XDR]..

3.1.6.4.1 Request Message

The *MyHealth@EU Dispensation Service* discard() request is initiated by an HP in the country of care (country B) for deleting a previously transmitted eDispensation document at the patient's country of affiliation (country A).
The *MyHealth@EU Dispensation Service* Discard message is a specialisation of the *IHE Provide And Register DocumentSet* transaction (ITI-41) request message as profiled in [IHE XDR]. The fields defined for the *ProvideAndRegisterDocumentSetRequest* message MUST be used as follow:

Element Name	MyHealth@EU Usage Convention
lcm:SubmitObjectsRequest	Container that can be used to provide the metadata for the transmitted discarded document, the submission set and the associations between documents (see below).
rim:RegistryObjectList	Container that contains (pointers to) the eDispensation discarded document
rim:ExtrinsicObject	For each eDispensation document a single extrinsic object MUST be defined. There MUST be a 1:1 id-correspondence between <rim:ExtrinsicObject> elements and <ihe:Document> elements. For a list of further metadata to be provided with an eDispensation discarded document see the table below.
rimext:Document	base64 encoded data for the eDispensation documents being submitted to the service provider. The <rimext:Document/> element also includes the document id attribute (rimext:Document/@id) of type xsd:anyURI to match the document ExtrinsicObject id in the metadata and providing the necessary linkage.
rim:Association	For each pair of MyHealth@EU coded and source coded documents an ebRIM association MUST be defined (see below for details on the encoding).

For each eDispensation discarded document (either MyHealth@EU coded or source coded) the following set of metadata MUST be provided:

Slot Name	Binding	Slot Value

id	Attribute	Identifer of the document. This identifier MUST be the same for <rim:ExtrinsicObject/@id> and <ihe:Document/@id>.
mimeType	Attribute	MUST be "text/xml"
objectType	Attribute	MUST be set acc. to section 4.3.1.2 of [IHE IT TF 3]
status	Attribute	MUST be "urn:oasis:names:tc:ebxml-regrep:StatusType:Approved"
creationTime	rim:Slot	MUST be given for XDR compatibility. MAY be ignored by the service provider.
languageCode	rim:Slot	MUST be given for XDR compatibility. MAY be ignored by the service provider.
sourcePatientID	rim:Slot	MUST be of the same value as $XDSDocumentEntryPatientId (see below)
healthcareFacilityTypeCode	classification	MUST be provided for XCA compatibility and correct addressing. Value MUST be set to ISO 3166-1 alpha-2 country code of the addressed PN.
practiceSettingCode	classification	MUST be provided for XDR compatibility but MAY be ignored by the service consumer. Value
confidentialityCode	classification	MUST be provided for XDR compatibility but MAY be ignored by the service consumer. Value SHOULD be set to "N", as long as the Minimal Metadata Profile is not published.
XDSDocumentClassCode	classification	eDispensation Discard code ("DISCARD-60593-1") coded according to specification in ITI TF-2a: 3.18.4.1.2.3.4 Coding of Code/Code-Scheme. As classification scheme 2.16.840.1.113883.6.1 MUST be used even if no official LOINC code exists currently for this type of document (eDispensation Discard).
XDSDocumentFormatCode	classification	Format qualifier as defined in [CDA templates];
$XDSDocumentEntryPatientId	External identifier	Equals to the patient identifier that was provided by the *MyHealth@EU Identification Service* (encoded as HL7 v3 II data type)
$XDSDocumentUniqueId	External identifier	MUST refer to the OID of the CDA document that is included within the <ihe:Document> element.

3.1.6.4.2 Expected Actions

The *MyHealth@EU Dispensation Service* provider shall discard all registry objects and documents as identified in the request. It shall respond to a DiscardRequest message with a registry response message containing a success indicator.
The MyHealth@EU Dispensation Service provider MUST verify that the requesting service user has sufficient rights to request a discard eDispensation operation for the identified patient. It MUST verify that the eDispensation was issued by the same HCPO

that now wants to discard it.
In case of an error that relates to the transmission of the request or the processing of the MyHealth@EU security token, the *MyHealth@EU Dispensation Service* provider MUST respond with a fault message according to section 4.4 of this document.

3.1.6.4.3 Response Message (Full Success Scenario)

If the *MyHealth@EU Dispensation Service* provider is able to decode the received dispensation document IDs and to properly process the request, it responds with an *ebXML Registry Response* with its status set to "urn:oasis:names:tc:ebxml-regrep:ResponseStatusType:Success"

```
<rs:RegistryResponse status="urn:oasis:names:tc:ebxml-regrep:ResponseStatusType:Success">
  …
</rs:RegistryResponse>
```

3.1.6.4.4 Response Message (Failure or Partial Failure Success Scenario)

If the *MyHealth@EU Dispensation Service* provider is able to decode the received dispensation document IDs but the deprecating of the dispensations failed, it responds with an *ebXML Registry Response* that contains a respective status indicator (see below). The response MUST contain a RegistryErrorList element that indicates the failure condition.
If none of the eDispensations was deprecated successfully, the response status MUST be set to "urn:oasis:names:tc:ebxml-regrep:ResponseStatusType:Failure". If at least one eDispensation was deprecated successfully, the response status MUST be set to "urn:ihe:iti:2007:ResponseStatusType:PartialSuccess".
A failure location MUST be provided if the error does not apply to all to-be-deprecated eDispensation documents. It MUST NOT be given if the error applies to all documents that are to be deprecated.
The severity of each registry error message MUST be set to "urn:oasis:names:tc:ebxml-regrep:ErrorSeverityType:Error". Multiple registry error messages MAY be included within a single <rs:RegistryErrorList> element. In extension to the XDS-b error messages defined in Table 4.1-11 of [IHE ITI TF-3] the error codes from Appendix 6.4 are defined for MyHealth@EU:

3.1.6.4.5 Response Message (Full Success Scenario)

The following example shows a possible positive response to the request given in section 3.4.1.2:

```
<?xml version="1.0" encoding="UTF-8"?>
<soapenv:Envelope xmlns=...>
<soapenv:Header>...</soapenv:Header>
<soapenv:Body>
      <rs:RegistryResponse
          status="urn:oasis:names:tc:ebxml-regrep:ResponseStatusType:Success">
       </rs:RegistryResponse>
</soapenv:Body>
</soapenv:Envelope>
```

The following example shows a possible negative response to the request given in section 3.4.1.2:

```
   <?xml version="1.0" encoding="UTF-8"?>
   <soapenv:Envelope xmlns=...>
   <soapenv:Header>...</soapenv:Header>
<soapenv:Body>
<rs:RegistryResponse
                  status="urn:oasis:names:tc:ebxml-regrep:ResponseStatusType:Failure">
          <rs:RegistryErrorList>
```

```
            <rs:RegistryError
                  severity="urn:oasis:names:tc:ebxml-regrep:ErrorSeverityType:Error"
                  errorCode="...." codeContext="No Match" location="1.42.20100103225206.3.3" />
            </rs:RegistryErrorList>
         </rs:RegistryResponse>
</soapenv:Body>
</soapenv:Envelope>
```

3.1.6.5 Security Audit Considerations

The service consumer MUST write an audit trail entry according to the *HP Assurance Audit Schema*. The service provider MUST write an audit trail entry according to the *Patient Privacy Audit Schema*.
The following table defines which categories MUST be filled (R), which MAY be filled (O) and which categories MUST NOT be used (X).

MyHealth@EU Instance	Opt.	Description
Event	R	Audited event
Requesting Point of Care	R / X	HCPO that issued the original request. This category MUST be filled by the service consumer. It MUST NOT be provided by the service provider.
Human Requestor	R	HP that triggered the request
Source Gateway	R	Service consumer node address at the country of Care
Target Gateway	R	Service provider node address at the country of the patient's affiliation
Audit Source	R	Legal entity that ensures the uniqueness of the identifiers that are used to identify active participants
Patient	R	Patient
Event Target	R	Reference to the discarded document
Error Message	O	Only used in case that the request handling was not completed successfully

For the Event Target Category the following fields MUST be provided:

Field Name	Opt.	Value Constraints
ParticipantObjectTypeCode	R	MUST be "2" (System Object)
ParticipantObjectTypeCodeRole	R	MUST be "4" (Resource)
ParticipantObjectDataLifeCycle	R	MUST be "14" (logical deletion)
ParticipantObjectIDTypeCode	R	MUST be "12" (URI)
ParticipantObjectID	R	MUST be string-encoded UUIDs of the discarded documents

3.1.6.6 Protocol Requirements

The *MyHealth@EU Dispensation Service* request and response messages will be transmitted using synchronous Web Services Exchange, according to the requirements specified in section 4.3 of this document.
Port types and bindings MUST be used as defined in the WSDLs given in sections 6.3 of this document. According to this the *MyHealth@EU Dispensation Service* operations' request and response data MUST be contained within the message body as follows:

MyHealth@EU Dispensation Service	Message Body

Put and Discard request	ProvideAndRegisterDocumentSet-b_Message (see section 6.3)
Put and Discard response	ProvideAndRegisterDocumentSet-bResponse_Message (see section 6.4.3)

MyHealth@EU Dispensation Service request messages MUST be protected by the service consumer (NCP-B) according to the MyHealth@EU message security considerations as defined in section 4.3.5.2 of this document. *MyHealth@EU Dispensation Service* response messages MUST be protected by the service provider (NCP-A) according to the MyHealth@EU message security considerations as defined in section 4.3.5.2 of this document.

3.1.7 MyHealth@EU Original Clinical Document Service



Figure 10 - OrCD Service Interface

The *MyHealth@EU Original Clinical Document Service* is used to share documents which faced no modification following its creation in the Country of Affiliation. Authorized Original Clinical Documents are:

- Laboratory results
- Hospital discharge reports
- Medical imaging reports (with or without access to referenced images
- Medical images

The implementation of the *MyHealth@EU Dispensation Service* is based on the following standards:

- ebRIM: OASIS/ebXML Registry Information Model v3.0 [OASIS ebRIM 3.0]
- ebRS: OASIS/ebXML Registry Services Specifications v3.0[12] [OASIS ebRS 3.0]
- XOP: XML-binary Optimized Packaging [W3C XOP]

and is compliant with the IHE profiles:

- XDR: IHE Cross-Enterprise Reliable Exchange [IHE ITI TF-1] [IHE ITI TF-2]

For discovery and localisation of the Original Clinical Document Service instance that is responsible for providing access to the identified patient's data see section 2.1.4 of this document.

3.1.7.1 List() Operation

The *MyHealth@EU Original Clinical Document Service* list() operation is implemented as IHE XCA Cross-community Access transaction. It is fully compliant with the ebRS 3.0 standard. The *MyHealth@EU Patient Service* list() operation

[12] The integration of ebRS and MTOM as used by MyHealth@EU is not compatible with the current version of OASIS ebRS.

includes the documents listed in the response meta-data, just like they would have been included in Cross-Gateway Fetch (SOAP 1.2 MTOP with XOP encoding attachments).

3.1.7.1.1 Request Message

The list() request is initiated by an HP in the country of care for retrieving the original clinical document of an identified patient. The respective request message builds upon the IHE XCA Cross-Community Access request message.

The <AdhocQueryRequest/> element that encapsulates the query parameters MUST be used as follows for MyHealth@EU:

Element Name	MyHealth@EU Usage Convention
ResponseOption/@returnComposedObjects	MUST be "true"
ResponseOption/@returnType	MUST be "LeafClassWithRepositoryItem" (XCF)
AdhocQuery	Container for holding the ebML stored query arguments. All arguments MUST be encoded as query slots (see table below).
AdhocQuery@id	MUST be "urn:uuid:f2072993-9478-41df-a603-8f016706efe8" which indicates a Fetch (which is an adaption of the findDocuments Query as defined in ITI TF-2a:3.18.1)

Only synchronous web services exchange MUST be used. The XDS Affinity Domain Option only applies to the national environment. Therefore it MUST NOT be used for NCP-2-NCP message exchange.

Stored query argument slots MUST be defined for the patient identifier and the document class code. The document format code and the document type code MAY be given. Other argument slots than the ones listed below MUST be ignored by the service provider and SHOULD NOT be issued by the service consumer.

Slot Name	Opt	Slot Value
$XDSDocumentEntryPatientId	R	Equals to the patient identifier that was provided by the *MyHealth@EU Identification Service* (encoded as HL7 v3 II data type)
$XDSDocumentEntryStatus	R	Only approved documents MUST be returned: 'urn:oasis:names:tc:ebxml-regrep: StatusType:Approved'
$XDSDocumentEntryClassCode	R	A LOINC code for the defined Original Clinical Documents must be used: • laboratory results – '11502-2' • hospital discharge reports – '34105-7' • medical imaging reports (with or without access to referenced images) – '18748-4' • medical images – 'x-clinical-image' As classification scheme 2.16.840.1.113883.6.1 MUST be used: e.g. for a laboratory result: '11502-2^^2.16.840.1.113883.6.1'

$XDSDocumentEntryTypeCode	O	A LOINC code for the defined Original Clinical Documents must be used: • laboratory results – '11502-2' • hospital discharge reports – '34105-7' • medical imaging reports (with or without access to referenced images) – '18748-4' • medical images – 'x-clinical-image' As classification scheme 2.16.840.1.113883.6.1 MUST be used: e.g. for a laboratory result: '11502-2^^2.16.840.1.113883.6.1'
$XDSDocumentEntryFormatCode	O	Format qualifier; see table below for details on applying these codes to the retrieval of a patient's original clinical documents. Only encodings of the original clinical documents that comply to the requested format code will be returned by the service provider. If this stored query slot is omitted the service provider MUST deliver all available encodings[13].

For the document format only the format codes listed in the following table MUST be used.

Document Format	Format Code	Document content
MyHealth@EU Original Clinical Document CDA document format	urn:eHDSI:orcd:pdf:2021	CDA-enveloped PDF/A encoding of the original document without any semantic transformation as source coded PDF with a CDA header per IHE XDS-SD.
MyHealth@EU Original Clinical Document PNG document format	urn:eHDSI:orcd:png:2021	Original Clinical document in PDF format.
MyHealth@EU Original Clinical Document JPEG document format	urn:eHDSI:orcd:jpeg:2021	Original Clinical document in JPEG format.

3.1.7.1.2 Example Request Message

The following excerpt from an MyHealth@EU Original Clinical Document Service list() request message shows a cross-NCP query request that contains argument slots for retrieving the laboratory report original clinical documents (LOINC code 11502-2) of an identified patient (patient identifier 90378912821). In this example the service consumer does not specify the requested encoding. Therefore the service provider MUST deliver all available encodings (e.g. CDA-enveloped PDF document).

```
<soapenv:Envelope xmlns:soapenv=" http://www.w3.org/2003/05/soap-envelope" xmlns:query="urn:oasis:names:tc:ebxml-regrep:xsd:query:3.0" xmlns:rim="urn:oasis:names:tc:ebxml-regrep:xsd:rim:3.0">
    <soapenv:Header /> <soapenv:Body>
    <query:AdhocQueryRequest>
        <query:ResponseOption returnComposedObjects="true" returnType="LeafClassWithRepositoryItem" />
            <rim:AdhocQuery id="urn:uuid:14d4debf-8f97-4251-9a74-a90016b0af0d">
```

13

```
            <rim:Slot name="$XDSDocumentEntryPatientId">
               <rim:ValueList>
                  <rim:Value>'90378912821^^^&amp;amp;1.3.6.1.4.1.21367.2005.3.7&amp;amp;ISO'</rim:Value>
               </rim:ValueList>
            </rim:Slot>
            <rim:Slot name="$XDSDocumentEntryStatus">
               <rim:ValueList>
                  <rim:Value>('urn:oasis:names:tc:ebxml-regrep:StatusType:Approved')</rim:Value>
               </rim:ValueList>
            </rim:Slot>
            <rim:Slot name="$XDSDocumentEntryClassCode">
               <rim:ValueList>
                  <rim:Value>(11502-2^^2.16.840.1.113883.6.1')</rim:Value>
               </rim:ValueList>
            </rim:Slot>
            <!-- Include associations whose sourceObject and targeObject attributes reference ExtrinsicObjects returned –
         </rim:AdhocQuery>
      </query:AdhocQueryRequest>
   </soapenv:Body>
</soapenv:Envelope>
```

3.1.7.1.3 Expected actions

The *MyHealth@EU Original Clinical Document Service* provider shall respond to a ListRequest message with the ListResponse message containing:

- the identified patient's original clinical documents together with a status notification (full success scenario, see section 3.3.1.4) or
- an error message (no patient summary provided, see section 3.3.1.5).

The *MyHealth@EU Original Clinical Document Service* provider MUST verify that the requesting service user has sufficient rights to access the original clinical document of the identified patient.
In case of an error that relates to the transmission of the request or the processing of the MyHealth@EU security token, the *MyHealth@EU Original Clinical Document Service* provider MUST respond with a fault message according to section 4.4 of this document.

3.1.7.1.4 Response Message (Full Success Scenario)

The MyHealth@EU list() response contains all available original clinical documents available for the patient that correspond with the provided classCode. The respective message builds upon the IHE XCA Cross-Community Access response and Cross-Community Access response messages.

The fields defined for the MyHealth@EU ListResponse message MUST be used as follows:

Element Name	MyHealth@EU Usage Convention
Query:AdhocQueryResponse	Response message according to IHE XCA Cross-Gateway Access response message [IHE XCA]
@status	For the full success scenario the response status MUST be set to "urn:oasis:names:tc:ebxml-regrep:ResponseStatusType:Success" or "urn:ihe:iti:2007:ResponseStatusType:PartialSuccess" (for details see table below)
rs:RegistryErrorList	In case that a warning is given by the service provider, this element holds the respective warning codes and messages. It must be used according to section 4.1.13 of [IHE ITI TF-3]
rim:RegistryObjectList	This element MUST be provided for the full success scenario. It MUST
rim:ExtrinsicObject	For each instance of an original clinical document a <rim:ExtrinsicObject/> element MUST be provided. Each <rim:ExtrinsicObject/> element is described and classified by metadata according to Table 4 below.
rimext:Document	This element MUST appear as the last element child of an <rim:ExtrinsicObject/> element. It may appear zero or one times. This element contains the base 64 encoded content of the document. The document contents are associated with the DocumentEntry (ExtrinsicObject) metadata by the fact that it is nested inside it within the XML. The base64 encoded document content MAY be encrypted. How encryption is applied and how the encryption key is negotiated should be subject to an additional specification on advanced security safeguards.

Each provided original clinical document MUST be further classified by metadata. The following table lists the usage conventions that MUST be followed for the MyHealth@EU Patient Service response message. If not stated otherwise the classification schemes as defined in section 4.3.1.2 of [IHE ITI TF-3] MUST be used. If no restrictions on metadata values are given, the metadata elements MUST be used as per [IHE XCA].

3.1.7.1.4.1 Common metadata elements

Metadata (ebRIM names)	Binding	Opt.	MyHealth@EU usage convention
status	Attribute	R	MUST be "urn:oasis:names:tc:ebxml-regrep:StatusType:Approved"
mimeType	Attribute	R	MUST be "text/xml" for all original clinical documents, since they are all embedded in a CDA document.
Name	Main	R	Must be provided and contain the name of the document.
Description	Main	O	Can contain additional information about the document

VersionInfo	Main	R	MUST be "1"
creationTime	rim:Slot	O	MAY be omitted by the service provider and MAY be ignored by the service consumer. If given, the value MUST be encoded as HL7 v2 Date Time "YYYY[MM[DD[hh[mm[ss]]]]]"
hash	rim:Slot	O	SHOULD be omitted by the service provider and MUST NOT be processed by the service consumer.
languageCode	rim:Slot	O	SHOULD be omitted by the service provider and MUST NOT be processed by the service consumer.
repositoryUniqueId	rim:Slot	O	MAY be omitted by the service provider and MAY be processed by the service consumer in an IHE-compatible NI-scenario.
serviceStartTime serviceEndTime	rim:Slot	O	Can contain the effectiveTime of the event.
size	rim:Slot	O	Can contain the size of the document.
sourcePatientId	rim:Slot	R	MUST contain the same value as XDSDocumentEntry.PatientId (see below).
sourcePatientInfo	rim:Slot	X	MUST NOT be used. Future versions of MyHealth@EU MAY define different protection levels for metadata and documents. Therefore all metadata elements that might carry medical or social information MUST be omitted.
classCode	Classification	R	laboratory results – '11502-2' hospital discharge reports – '34105-7' medical imaging reports (with or without access to referenced images) – '18748-4' medical images – 'x-clinical-image' As classification scheme "urn:oid:2.16.840.1.113883.6.1" MUST be used
typeCode	Classification	O	laboratory results – '11502-2' hospital discharge reports – '34105-7' medical imaging reports (with or without access to referenced images) – '18748-4' medical images – 'x-clinical-image' As classification scheme "urn:oid:2.16.840.1.113883.6.1" MUST be used Could be used for more fine-grained description of the type of original coded element.
author	Classification	O	Can be used optionally for laboratory results, hospital discharge reports and medcial imaging reports and medical images.

confidentialityCode	Classification	R	MUST be provided for XCA compatibility but MAY be ignored by the service consumer. Value SHOULD be set to "N", as long as the Minimal Metadata Profile is not published.
formatCode	Classification	R	MUST be “urn:eHDSI:orcd:pdf:2021" for CDA-enveloped PDF document. In the case the embedded document format is PNG or JPEG, the formatCode must be “urn:eHDSI:orcd:png:2021” or “urn:eHDSI:orcd:jpeg:2021”
XDSDocumentEntry.uniqueId	ExternalIdentifier	R	MUST hold the OID of the document. The document unique id value MUST be the same as the value of the document’s <ClinicalDocument/id> CDA header element.
XDSDocumentEntry.patientId	ExternalIdentifier	R	MUST hold the patient identifier. The service consumer MUST verify that this id matches the patient Id that was discovered by the MyHealth@EU Identification Service.

Table 1: MyHealth@EU Original Clinical Document Common Metadata

3.1.7.1.4.2 Metadata elements specific to laboratory results

Metadata (ebRIM names)	Binding	Opt.	MyHealth@EU usage convention
Author/authorSpeciality	Attribute	O	Can be provided by the service provider. If possible the specialty text must be provided.

Table 2: MyHealth@EU Original Clinical Document Laboratory Results Metadata

3.1.7.1.4.3 Metadata elements specific to hospital discharge reports

Metadata (ebRIM names)	Binding	Opt.	MyHealth@EU usage convention
Author/authorSpeciality	Attribute	O	Can be provided by the service provider. If possible the specialty text must be provided.
eventCodeList	Classification	O	ClassficationScheme: urn:uuid:2c6b8cb7-8b2a-4051-b291-b1ae6a575ef4 Can be used to contain the reason of hospitalisation. If possible the reason of hospitalisation COULD be added in textual form.

Table 3: MyHealth@EU Original Clinical Document Hospital Discharge Reports Metadata

3.1.7.1.4.4 Metadata elements specific to medical imaging reports and medical images

Metadata (ebRIM names)	Binding	Opt.	MyHealth@EU usage convention

Author/authorSpeciality	Attribute	O	Can be provided by the service provider. If possible the specialty text must be provided.
eventCodeList	Classification	O	ClassficationScheme: urn:uuid:2c6b8cb7-8b2a-4051-b291-b1ae6a575ef4 Used to contain the reason of hospitalisation. If possible the reason of hospitalisation COULD be added in textual form.
author	Classification	O	ClassficationScheme: urn:uuid:93606bcf-9494-43ec-9b4e-a7748d1a838d Used to contain the requester specialty. If possible the requester specialty COULD be added in textual form.

Table 4: MyHealth@EU Original Clinical Document Medical Imaging Reports and Medical Images Metadata

Other metadata than the ones listed above SHOULD NOT be provided by the service provider and MUST NOT be processed by the service consumer.
By definition, several original clinical documents can be provided per patient.

If a warning is to be transmitted to the HP (see section 3.1.4) the ebXML Registry Error mechanism MUST be used with a syntax as defined in section 3.43.5 of [IHE ITI TF-2]. As the location of the warning is implied, the respective location attribute SHOULD be empty.

3.1.7.1.5 Response Message (No original clinical documents provided)

If the *MyHealth@EU Original Clinical Document Service* provider is unable to respond with documents of the requested type, in the requested encoding it MUST respond with a ListResponse message that only contains a <AdhocQueryResponse/RegistryResponse> element.

For a full list of error messages defined for IHE X* see table 4.1-11 in [IHE ITI TF-3]. The table from Appendix 6.4 lists the additional, MyHealth@EU-specific response status types and error/warning/info codes to be used within the <RegistryErrorList> element.

3.1.7.1.6 Example Response Message

The following message is a possible response to the sample request message given in section 3.3.1.2. The patient's country of affiliation responds with both encodings.

3.1.7.2 Security Audit Considerations

The service consumer MUST write an audit trail entry according to the HP Assurance Audit Schema. The service provider MUST write an audit trail entry according to the Patient Privacy Audit Schema.
The following table defines which categories MUST be filled (R), which MAY be filled (O) and which categories MUST NOT be used (X).

MyHealth@EU Instance	Opt.	Description
Event	R	Audited event
Requesting Point of Care	R / X	HCPO that issued the original request. This category MUST be filled by the service consumer. It MUST NOT be provided by the service provider.
Human Requestor	R	HP that triggered the request
Source Gateway	R	Service consumer node address at the country of Care
Target Gateway	R	Service provider node address at the country of the patient's affiliation
Audit Source	R	Legal entity that ensures the uniqueness of the identifiers that are used to identify active participants
Patient	R	Patient
Event Target	R	Subject to the Query
Error Message	O	Only used in case that the request handling was not completed successfully

For the Event Target Category the following fields MUST be provided:

Field Name	Opt.	Value Constraints
ParticipantObjectTypeCode	R	MUST be "2" (System Object)
ParticipantObjectTypeCodeRole	R	MUST be "24" (Query)
ParticipantObjectIDTypeCode	R	MUST be "10" (Search Criteria)
ParticipantObjectID	R	MUST be string-encoded UUIDs of the returned documents

3.1.7.3 Protocol Requirements

The *MyHealth@EU Patient Service* List() request and response messages will be transmitted using synchronous Web Services Exchange, according to the requirements specified in section 4.3 of this document.
Port types and bindings MUST be used as defined in the WSDL given in section 6.4.2 of this document. Acc. to this the MyHealth@EU Patient Service List() operation's request and response data MUST be contained within the message body as follows:

MyHealth@EU OrCD Service	Message Body
List request	CrossGatewayQueryRetrieve_Message (see section 6.4.2)
List response	CrossGatewayQueryRetrieveResponse_Message (see section 6.4.2)

The request message MUST be protected by the service consumer (NCP-B) according to the MyHealth@EU message security considerations as defined in section 4.3.5.2 of this document. The response message MUST be protected by the service provider (NCP-A) according to the MyHealth@EU message security considerations as defined in section 4.3.5.2 of this document.

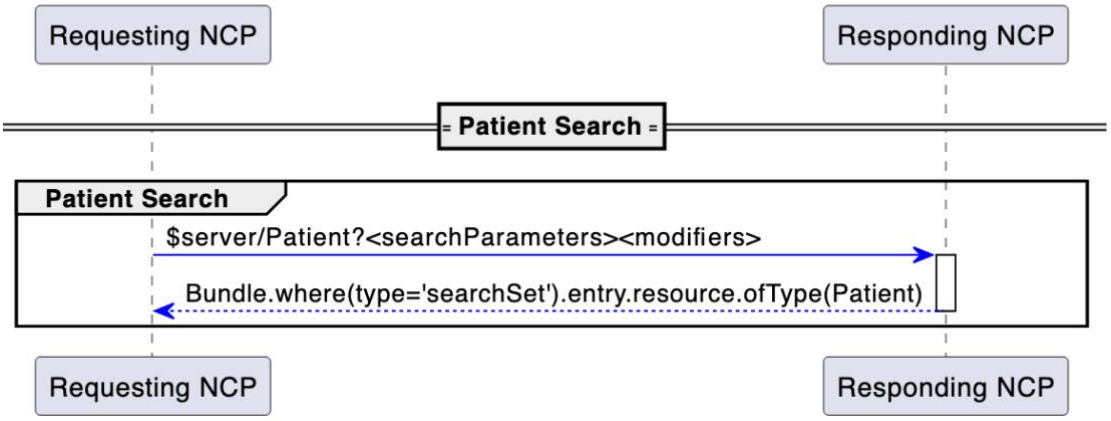
3.2 HL7 FHIR-based Services

3.2.1.1 MyHealth@EU Patient REST API

Similarly to the IHE-based Identification Service uncovers a valid patient. In this case, the FHIR standard is using parameters like business identifier, name, birth date and gender.

3.2.1.1.1 Search request

The request is done using REST convention through HTTP protocol. The message exchanged is passed in a JSON file.



Search parameters

Parameter	Optionality	Type
Business identifier	Mandatory	token
Name	Optional	string
Birth date	Optional	date
Gender	Optional	token

3.2.1.1.2 Examples

All the Patient having this identifier:

GET $server/Patient?identifier=http://identifiers.mycountry|123456789

All the Patient having in the name “doe” born the 1985-01-14 and of male gender:

GET $server/Patient?name=doe&birthdate=eq1985-01-14&sex=male

Response – returned patient data

Parameter	Optionality
Business identifier	Mandatory
Name	Mandatory

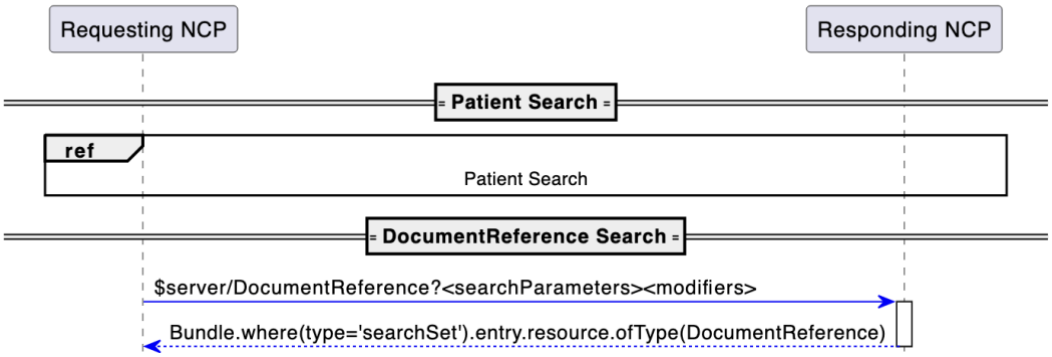
Birth date	Mandatory
Gender	Mandatory

3.2.1.2 MyHealth@EU DocumentReference REST API

In order to retrieve a lab results report (LRR) for a patient, a REST call must be done. The document in returned in the form of a FHIR Bundle compliant with the DocumentReferenceMyHealthEu profile.

3.2.1.2.1 Search request

Using the identifier of the patient, it is possible to get all lab results reports for this particular patient.



3.2.1.2.2 Search parameters

Parameter	Optionality	Type
Document type	Mandatory	token
Type of content	Optional	token
Study type	Optional	token
Patient business identifier	Mandatory	reference
Date	Optional	date

The document type codes that can be used in the search query are the following:

34105-7	Hospital Discharge Report
11502-2	Laboratory Result
18748-4	Medical Imaging Report
11369-6	Immunizations history (EUVAC)

Example
All the lab reports of the Czech patient 456789123 created from 2023, Feb 20th :

GET $server/DocumentReference?type=http://loinc.org|11502-2&patient.identifier=https://ncez.mzcr.cz/standards/fhir/sid/rid|456789123&date=ge2023-02-20

All the hospital discharge reports of the Czech patient 456789123 of type PDF.

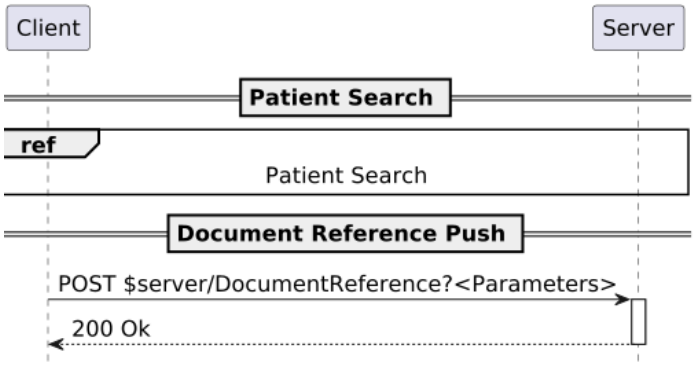
GET
$server/DocumentReference?type=http://loinc.org|34105&patient.identifier=https://ncez.mzcr.cz/standards/fhir/sid/rid|456789123&contenttype=application/pdf

Fetch the vaccination history for Czech patient 456789123:

GET $server/DocumentReference?type=http://loinc.org|11369-6&patient.identifier=https://ncez.mzcr.cz/standards/fhir/sid/rid|456789123

3.2.1.2.3 Push request

Using the identifier of the patient, it is possible to the country of treatment to send a document reference to the country of affiliation.



The only FHIR flow that is using the document reference push is the Hospital Discharge Report.

3.2.1.2.4 Push parameters

Parameter	Optionality	Type
Patient business identifier	Mandatory	reference
Payload (request body)	Mandatory	DocumentReference

Example:

Pusshing an HDR Report from patient 89121210976 to country of affiliation

POST
$server/DocumentReference?patient.identifier=https://www.ehealth.fgov.be/standards/fhir/core/NamingSystem/ssin|89121210976

Payload: A Json formatted DocumentReference resource in the body of the request

For more information on how to implement an HDR push please visit
https://fhir.ehdsi.eu/hdr/

3.2.1.3 MyHealth@EU Bundle REST API

In order to retrieve a specific lab result for a patient, a REST call must be done by passing the identifier of the requested bundle. The requester and patient information is passed in a collection of assertions that is sent inside a JWT Token. The document is returned in the form of a FHIR Bundle compliant with the DocumentReferenceMyHealthEu profile.
The JWT token MUST contain a valid TRC claim linking the patient identifier to the HP, equivalent to the TRC assertion requirement in the IHE service.
The Bundle REST API also allows the user to send Bundle information to the server. This is currently used to send the Hospital Discharge Report information to the country of origin of the patient.

3.2.1.3.1 Search parameters

Parameter	Optionality	Type
Bundle Identifier	Mandatory	string

Example

Retrieve SimpleChemistryResultReport Bundle for of the Czech patient 456789123:

GET $server/Bundle/SimpleChemistryResultReport (patient id is sent in the TRC assertion encaspsulated inside a JWT token)

4 MyHealth@EU Communication and Messaging Infrastructure

MyHealth@EU service providers and consumers use the MyHealth@EU messaging infrastructure to exchange request and response messages among each other. The message infrastructure builds upon the MyHealth@EU communication infrastructure that connects the MyHealth@EU network of trusted nodes (Figure 13).

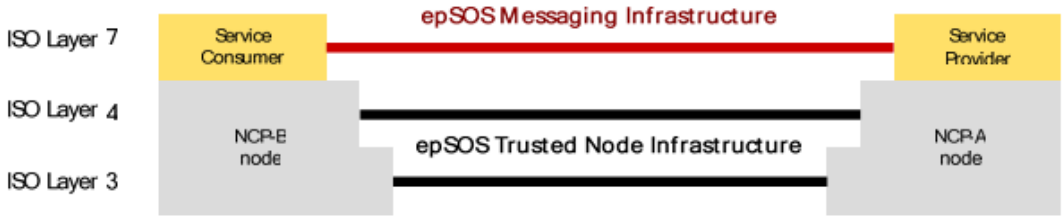


Figure 13: MyHealth@EU Trusted Nodes and Messaging Infrastructure

The MyHealth@EU trusted node infrastructure implements the core MyHealth@EU security services that ensure the confidentiality of medical data transmission and the availability and authenticity of MyHealth@EU services:

- physical private network (Testa NG),
- message encryption (TLS) and integrity protection, and
- mutual NCP authentication.

The MyHealth@EU messaging infrastructure provides mechanisms for the implementation of the derived MyHealth@EU security services (e. g. non-repudiation and access control) and for the standardised enveloping of data and documents:

- transmission of authenticated HP attributes,
- common message format, and
- signature on message elements for auditing and brokering of document authenticity claims.

It must be noted that only NCPs acting as MyHealth@EU service providers and consumers are part of the MyHealth@EU trusted node infrastructure as specified in this document. Points of Care within country B or national data registries/repositories in country A have to be connected to the MyHealth@EU trusted nodes by means that respect the MyHealth@EU end-to-end privacy, security and data protection requirements. See [Section I - Security Policies] and [Security Services Specification] for details.

4.1 Audit Trail Implementation

The mutual trust between a service consumer and a service provider is based on a mutually trusted, secure channel between the underlying network nodes.

The establishment of mutual trust between nodes is performed by:

- IPSec [RFC 4301]
- Transport Layer Security v1.2 [RFC 5246]
- IHE Audit Trail and Node Authentication [IHE ITI TF-2a]

4.1.1 TLS Configuration

All network nodes running MyHealth@EU service consumers or service providers MUST be implemenented as IHE *Secure Node* actors acc. to the IHE ATNA profile. The establishment of mutual trust and the setup of the secure transport layer channel between two MyHealth@EU nodes are always initiated by a service consumer that connects to a service provider.

The messages for the establishment of the basic transport layer secure channel correspond to the TLS handshake protocol as profiled in the IHE ATNA Integration profile (transaction ITI-19 as specified in section 3.19 of [IHE ITI TF 2a]).

With respect to the ITI-19 transaction specification the following constraints and extensions apply:

- Algorithms and key lengths MUST be used acc. to [Cryptographic Algorithms].
- The node certificates MUST comply with the MyHealth@EU Node Authentication Certificate Profile
- [X.509 Certificate Profiles]
- The issuing CA and all components and services for managing the lifecycle of the MyHealth@EU Node Authentication Certificates must comply with the respective MyHealth@EU security policies (see [Security Services Specification]).

4.2 Time Synchronisation

Time synchronisation within the network of MyHealth@EU gateways is performed by:

- Network Time Protocol (Version3) [RFC 1305] as described in the
- IHE Consistent Time Integration Profile [IHE ITI TF-2a]

Stratum 2 time servers (*Consistent Time Mediators)* SHOULD by operated by NCPs. All services that rely on consistent time within the MyHealth@EU Circle of Trust MUST be operated on a node that acts as a stratum 3 time server (*Consistent Time Consumer)*. This in particular holds for

- services that apply or verify digital signatures on messages, medical data or assertions,
- services that contribute to the security audit trail.

MyHealth@EU time synchronisation for *Consistent Time Consumers* is handled by

respective mechanisms of the underlying operating systems. The messages exchanged correspond to the NTP transactions described in detail in RFC 1305 and http://www.ntp.org. The underlying network protocol is UDP (port 123). Authentication MAY be enabled with the *ntp authenticate* command
Consistent Time servers that represent a Stratum n+1 server SHOULD have a configuration with a default polling interval of 4096 seconds at a minimum in order to synchronize the MyHealth@EU reference time to all nodes. Following time servers SHOULD only configure a polling interval of 65536 seconds.
Each Consistent Time Mediator SHOULD accept a maximum clock skew of 256 seconds. With respect to lower the system resources (due to incoming requests) of the Consistent Time Source, Consistent Time Mediators SHOULD peer themselves.

4.3 MyHealth@EU Common Message Format

The MyHealth@EU Common Message Format defines the structure and characteristics of the messages exchanged through MyHealth@EU and establishes the preconditions for successful communication. The MyHealth@EU Common Message Format describes only the structure of messages as they flow between service consumers and service providers. Messages within NCP's or national infrastructures may use any format desired, and need to be translated to the MyHealth@EU Common Message Format before transmission to another NCP.

4.3.1 Transport Layer Profile

All messages MUST be sent over HTTP 1.1 connections that are layered on top of the MyHealth@EU trusted node infrastructure (see section 4.1).

4.3.2 Message Layer Profile

The MyHealth@EU Common Message Format is a SOAP 1.2 [W3C SOAP 1.2] message contained as the body of an HTPP 1.1 [RFC 2616] message.
All messages MUST be SOAP Envelopes with an XML payload in the SOAP Body. Optional binary data MUST be carried as Base 64 encoded octets within the XML payload if not otherwise stated for the respective operations. Request messages MUST be sent using an HTTP POST, response messages are carried over the backchannel, i.e. the HTTP response.
The encoding of the containing XML document MUST be set to UTF-8[14].
All MyHealth@EU SOAP messages MUST comply with the WS-I Basic Profile 1.1[15] [WSI BP 1.1].
All MyHealth@EU SOAP messages MUST comply with the WS-I Basic Security Profile 1.1 [WSI SBP 1.1].

4.3.3 XML Message Schema Format

All MyHealth@EU SOAP message MUST be described in a WSDL 1.1 [W3C WSDL 1.1] Service Description.
All WSDL type definitions MUST be in XML Schema format. One Schema must be provided for the request message, and one for the response message.
For better maintainability, an XML Schema import or include statement in the WSDL file SHOULD be used, so the XML Schema can be maintained and reused as a separate entity.

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[14] UTF-8 is more efficient than UTF-16 for European languages. Older encodings such as ISO-8859-x do not cover all languages in a single encoding, and will only pose interoperability problems. UTF-8 is the default in XML, and coverage is a requirement of the XML specification.

[15] While WSI Basic profile 1.1 does not formally support SOAP 1.2, it takes into consideration SOAP 1.2 by having requirements which are specifically for compatibility with SOAP 1.2. IHE and MyHealth@EU plan to consider adoption of WS-I BP 2.0 [WSI BP 2.0] as soon as it is approved by WSI.

If the XML Schema is small, and reuse is not expected, the entire Schema MAY be specified in the WSDL types section (especially in the case of RPC-style transactions where one or a few parameters of rather simple types are used).

4.3.4 SOAP Binding

For all messages a SOAP 1.2 HTTP Binding MUST be provided in the WSDL. All SOAP Bindings in the WSDL MUST specify style="document".
All SOAP Bindings in the WSDL MUST specify use="literal".
The naming of the messages MUST be as defined for the used standard.
The "soapenv:mustUnderstand="1"" attribute MUST be set as defined for the used standard.

4.3.5 Embedding of Security Token

Each SOAP message MUST include a <wsse:Security> section within the SOAP header.

4.3.6 SAML Assertions

Request messages are safeguarded by up to two SAML assertions that attest the authenticity of the user and the existence of a treatment relationship (See chapter 2 on which assertions are required for each operation).
SAML assertions are contained within the <wsse:Security> section of the SOAP header.
The HP Identity Assertion always takes the role of a Supporting Token.
The saml:Advice element MUST be used to define the linkage between the two assertions (see [SAML Profile]). Both assertions MUST have been issued for the same subject. The relying party MUST verify the correct linkage of the SAML assertions and the match of the <Subject> elements' contents.

4.3.7 Message Signature

To preserve the integrity and authenticity of a message and to attest the NCP origin of the message, elements of each message MAY be signed by the protection token. If WS SecurityPolicy is used, a signed elements assertion MUST be used to refer to the message parts to be signed[16].
The following table defines which token MUST be used as protection token and which elements of the message MUST be covered by the signature.

	Request Message	**Response Message**
Protection Token	X.509 Token of NCP-B	X.509 Token of NCP-A
Signed Elements	/Envelope/Body /Envelope/Header/Security/Assertion	/Envelope/Body

Signatures MUST be placed within a XML-Signature compliant <ds:Signature/> element inside the SOAP security header. The recommendations given in section 8 of [OASIS WS-Security 1.1] SHOULD be considered. In addition the following constraints apply:

Signature Parameter	**Usage Convention**
CanonicalizationMethod	SHOULD be "http://www.w3.org/2001/10/xml-exc-c14n#"
Transformation	Exclusive XML canonicalization SHOULD be used (http://www.w3.org/2001/10/xml-exc- c14n#, acc. [W3C XMLDSig] and [W3C XML-EXC 1.0]). As inclusive namespaces other prefixes than the ones defined in section 3.1.6 of this document MUST NOT be used.

[16] Only in cases where the framework does not allow for multiple XPath expressions, a signed parts assertion SHOULD be used.

SignatureMethod	The signature method MUST comply with the MyHealth@EU recommendations on algorithms and key lengths (see [Cryptographic Algorithms]). For signing message elements the signature method • http://www.w3.org/2001/04/xmldsig-more#rsa-sha256 SHALL be used.
DigestMethod	The hash algorithm MUST comply with the MyHealth@EU recommendations on algorithms and key lengths (see [Cryptographic Algorithms]). For signing message elements the digest method • http://www.w3.org/2001/04/xmlenc#sha256 SHALL be used.
KeyInfo	This element MUST contain a wsse:SecurityTokenReference element which references the protection token.

4.3.8 Processing of SOAP Messages

A sending service consumer SHOULD add SOAP Message Headers in the order in which the receiving service provider is expected to process them.
A receiving service provider MUST not rely on a specific order of SOAP Message Headers for correct processing.
A receiving service provider MAY rely on a specific order of SOAP Message Headers for faster processing.

4.4 Exception Handling

In general MyHealth@EU distinguishes between four coarse grained failure situations that MAY require an exchange of respective fault messages between NCPs:

- Communication failures: The message cannot be delivered to the designated service; e.g. because the establishment of a secure communication link failed or a link is broken.
- MyHealth@EU Message encoding and consistency failures: The message was received but it cannot be processed; e.g. because the security token is broken or the message does not comply with the MyHealth@EU message encoding and security rules
- Message processing failures: The message was decoded but it cannot be processed; e.g. because of security/privacy reasons or because the request does not match with the states of the affected security and business objects
- Document encoding failures: The message can be processed but the requested document cannot be completely encoded as specified in [CDA templates].

General system failures are handled based on their implication (e.g. if a message cannot be delivered due to a buffer error, the communication failure mechansim is used to propagate the error; if the same failure occurs during message processing, the message processing failure mechanism is used).
In the following sections it is defined, how failures of the first three kinds are handled by MyHealth@EU. Failures of the fourth kind are handled within the documents (e.g. by nullifying fields or using specific error codes; see [CDA templates]).

4.4.1 Communication Failures

Communication failures are raised by the existing mechanisms of the communication and messaging protocols. They are handled on the layer where they occurred. MyHealth@EU does not define new error codes for these kinds of failures and MyHealth@EU does not

define any requirements for raising and processing these errors that are beyond the presetting of the respective standards. This includes that errors of this kind can occur at both communicating gateways and MAY require action to be taken by both gateways.
In case of a communication failure an audit trail entry MUST be written at NCP-B:

MyHealth@EU Instance	Opt.	Description
Event	R	Service that was to be called
Requesting Point of Care	R	HCPO that issued the original request.
Human Requestor	R	HP that triggered the request
Source Gateway	R	Service consumer node address at the country of Care
Target Gateway	R	Service provider node that did not respond to the request
Audit Source	X	-
Patient	R	Patient
Event Target	X	-
Error Message	R	Error data as provided by the layer that detected the communication failure

4.4.2 Encoding and Consistency Failures

Encoding and consistency problems are detected at the protocol terminator or other MyHealth@EU-side internal components at the service providing NCP. Errors that origin in an improper encoding of the message (envelope, header) or in an inaccurate use of security objects are covered by the SOAP error mechanism.

4.4.3 SOAP Error Profile

Information on faults that occurred during the processing of a request are placed into a SOAP response message body as SOAP 1.2 faults. The respective data type MUST be instantiated as defined in [W3C SOAP 1.2]. MyHealth@EU specific error information is encoded with the following elements:

Element name	Format	Opt.	Content
Code/Subcode/Value	QName	R	MyHealth@EU error code; see tables below for the defined values
Reason/Text	QName	R	Description of the error (by default the error code is used as the error description; nevertheless a NCP implementation MAY provide the error condition (see tables below) or even more detailed information on the reason of the failure in this element). In cases where it is possible a document identifier should be
Detail	Element	O	Element with tag named after the exception that was thrown and the reason of the failure in the contents
Node	URI	O	URI of the system component that caused the failure or (URI encoded) OID of the object that caused the error. The semantics of this entry MUST be determinable by the error code. By default this element holds the URI of the Service Provider where the error was detected.

Table 11: Usage conventions for SOAP faults

In case the <detail/> element exists the receiver of the fault message MUST NOT process the <detail/> element but SHOULD dump its contents into the respective field of the audit trail entry.

4.4.4 General Message Handling Errors

General message handling faults are detected upon receipt of a message. Usually they are not specific for a certain transaction and usually origin in weaknesses related to the implementation, configuration and operation of the MyHealth@EU NCP.
The following table lists all general message handling errors. These errors MUST be handled acc. to the MyHealth@EU SOAP error profile (see section 4.4.2.1).

Condition	Code	Subcode	Action to be taken
The service provider is not able to fulfil the request due to an internal problem.	Receiver	Busy	Both NCPs MUST write an audit trail entry. The service requestor SHOULD send the request again.
The service provider cannot write an audit trail entry.	Receiver	Audit Log Failure	The service consumer MUST write an audit trail entry. The service provider MUST write a log entry to the systems log. The system administrator MUST process this failure because it indicates a mis-configuration or software error.

4.4.5 SOAP Message Encoding and Addressing Errors

Message encoding and addressing faults are detected upon receipt of a message or at the protocol terminator. Usually they are not specific for a certain transaction and usually origin in weaknesses related to the implementation, configuration and operation of the MyHealth@EU NCP.
The following table lists all message encoding and addressing errors. These errors MUST be handled acc. to the MyHealth@EU SOAP error profile (see section 4.4.2.1).

Condition	Code	Subcode	Action to be taken
The protocol terminator cannot decode the message because of a schema violation in the SOAP envelope	MustUnderstand or DataEncodingUnknown (depending on the source of the error)	Decoding Failure	No audit trails are written. The service consumer MUST write a log entry to the systems log. The system administrator MUST process this failure because it indicates a mis-configuration or software error.
The protocol terminator cannot validate a message because of an unknown namespace or schema	DataEncodingUnknown	Unknown Schema	No audit trails are written. The service consumer MUST write a log entry to the systems log. The system administrator MUST process this failure because it indicates a mis-configuration or software error.

The protocol terminator cannot process the message because it does not know or not support the requested service.	MustUnderstand or DataEncodingUnknown (depending on the source of the error)	Unknown Transaction	No audit trails are written. The service consumer MUST write a log entry to the systems log. The system administrator MUST process this failure because it indicates a mis-configuration or software error.
The protocol terminator rejects the message because of a ver- sion mismatch (e. g. service consumer uses deprecated ver- sion of the spec)	MustUnderstand or DataEncodingUnknown (depending on the source of the error)	Version Mismatch	No audit trails are written. The service consumer MUST write a log entry to the systems log. The system administrator MUST process this failure because it indicates a mis-configuration or software error.

4.4.6 Security Header Encoding and Consistency Errors

Security header encoding and consistency errors are detected by the security manager component. Usually they are not specific for a certain transaction and usually origin in weaknesses related to the implementation, configuration and operation of the MyHealth@EU NCP.
The following table lists all security header related errors. These errors MUST be handled acc. to the MyHealth@EU SOAP error profile (see section 4.4.2.1).

Condition	Code	Subcode	Action to be taken
The provided HP Identity Assertion does not contain all of the required attributes.	Sender	HP Missing Attributes	Both service consumer and service provider MUST indicate the failure of the security header validation in the audit trail entry of the service in which it used (XCPD, XCA, XDR, FHIR). The
The provided HP Identity Assertion is not valid or timed out.	Sender	Invalid Security Token	Both service consumer and service provider MUST indicate the failure of the security header validation in the audit trail entry of the service in which it used (XCPD, XCA, XDR, FHIR). The
The patient identifier is not valid.	Sender	Unknown Patient	Both service consumer and service provider MUST indicate the failure of the security header validation in the audit trail entry of the service in which it used (XCPD, XCA, XDR, FHIR). The HP at the country of care
An attesting (message) signature cannot be verified.	Sender or Receiver (depending on the source of the failure)	Invalid NCP Signature	Both service consumer and service provider MUST indicate the failure of the security header validation in the audit trail entry of the service in which it used (XCPD, XCA, XDR, FHIR). The service provider MUST write a log entry to the systems log. The system administrator MUST process this failure because it

The use of SHA-1 as a digesting method is not allowed.	Sender or Receiver (depending on the source of the failure)	Weak Digest	Both service consumer and service provider MUST indicate the failure of the security header validation in the audit trail entry of the service in which it used (XCPD, XCA, XDR, FHIR). The service provider MUST write a log entry to the systems log. The service consumer SHOULD re-issue the request using SHA-2 for digesting (message signature and assertion signatures). The service provider SHOULD use the same digesting method for message signatures as the service
The requestor provided a Confirmation Assertion which is not accepted by the service provider.	Sender	Weak Authorisation	Both service consumer and service provider MUST indicate the failure of the security header validation in the audit trail entry of the service in which it used (XCPD, XCA, XDR, FHIR). The HP SHOULD trigger the issuance

4.4.7 Audit Trail Considerations

In case of a general message handling error, NCP-B MUST write a full audit trail including an error section as defined in chapter 4.4.3.

NCP-A MUST fill all audit trail information that could be decoded from the request message.

If the requested operation cannot be decoded the Event Identification section of the HP Assurance Audit Schema MUST be used as follows:

Field Name	Value Constraints
EventID	MUST be set to EV(EHDSI-00, "unknown", unknown)
EventActionCode	MUST be set to E (execute).
EventDateTime	Time of the occurrence of the failure
EventOutcomeIndicator	According to DICOM MUST be "8" for temporal or recoverable failures and "12" for permanent failures.

If the HP identity cannot be decoded from the HP Identity Assertion, the Human Requestor section of the HP Assurance Audit Schema MUST be used as follows:

Field Name	Value Constraints
UserID	Subject and issuer MUST be set to "unknown".
UserName	MUST be set to "unknown"
UserIsRequestor	"true"
RoleIDCode	MUST be omitted.

If the patient identity cannot be decoded from the request, the Patient section of the HP Assurance Audit Schema MUST be used as follows:

Field Name	Value Constraints
ParticipantObjectTypeCode	MUST be "1" (Person)
ParticipantObjectTypeCodeRole	MUST be "1" (Patient)

ParticipantObjectIDTypeCode	EV(2, RFC-3881, "Patient Number")
ParticipantObjectID	MUST be "unknown"

5 MyHealth@EU Profiles on Assertions and Certificates

MyHealth@EU security mechanisms build upon SAML assertions and digital certificates as core security objects. This chapter provides the respective MyHealth@EU security object profiles on SAML and X.509.

5.1 Cryptographic keys and Algorithms

All cryptographic keys and algorithms used for MyHealth@EU and its implementations MUST fulfil at least the requirements of [SOG-IS-2020] for "Legacy" level of security with validity longer than end of 2025.. This corresponds to a minimum of 128-bit security (symmetric equivalent).

For more information on cryptographic keys and algorithms, please check [Cryptographic Algorithms].

5.2 HP Identity Assertion

The HP Identity Assertion is a profiled SAML v2.0 assertion. It has *Sender-Vouches* configured as the confirmation method.

For more information on HP Identity Assertion, please check [SAML Profile].

5.3 Treatment Relationship Confirmation Assertion

The Confirmation Assertion is a profiled SAML v2.0 assertion. It attests the existence of a treatment relationship between a patient and a HCPO and provides information about the context of a certain treatment scenario.

For more information on Confirmation Assertion, please check [SAML Profile].

5.4 MyHealth@EU Certificate Profiles

The following sections define how to set up MyHealth@EU compliant X.509 certificates. All certificates issued by a CA that are used for MyHealth@EU are compliant X.509 certificates.
For more information on X.509 certificates, please check [X.509 Certificate Profiles].

6 Appendix

6.1 Coding Conventions (Normative)

6.1.1 Country Codes

Fields carrying country code values MUST be coded in accordance with [ISO 3166-1] Alpha 2 codes.

Country Code (normative)
BE
BG
CZ

DK
DE
EE
IE
GR
ES
FR
HR
IT
CY
LV
LT
LU
HU
MT
NL
AT
PL
PT
RO
SL
SK
FI
SE

The following exceptions apply:

- For Greece the country codes "GR" and "EL" are allowed.

6.2 MyHealth@EU Identifiers (Normative)

6.2.1 Uniform Resource Names (URNs)

The following URNs are defined:

URN	Description
urn:epsos:names:wp3.4:subject:healthcare-facility-type	Signals that a SAML attribute refers to the kind of the healthcare facility assigned to the subject. See [SAML Profile] for details.

urn:epsos:names:wp3.4:subject:clinical-speciality	Signals that a SAML attribute refers to the kind of the clinical speciality of the subject. See [SAML Profile] for details.
urn:epsos:names:wp3.4:subject:on-behalf-of	Signals that a SAML attribute refers to the person who authorized the subject's access to MyHealth@EU services. See [SAML Profile] for details.

6.2.2 MyHealth@EU OIDS

The following OIDs are defined:

OID	Type	Values / Description
1.3.6.1.4.1.12559.11.10.1.3.2.**2.1**	Code list	{"AdditionalDemographicsRequested", "DemographicsQueryNotAllowed", "EHICDataRequested", "InsufficientRights", "PrivacyViolation", AnswerNotAvailable", PolicyViolation", "PatientAuthenticationRequired" }
1.3.6.1.4.1.12559.11.10.1.3.2.**2.2**	Code list	{"Hospital", "Resident Physician", "Pharmacy", "Other" }
1.3.6.1.4.1.12559.11.10.1.3.2.**3.1**	Code list	{ "eHDSI pivot" }
1.3.6.1.4.1.12559.11.10.1.3.2.**4.1**	Coded values	1: Opt-In Policy 2: Opt-Out Policy

6.2.3 MyHealth@EU CDA Documents and Codes

MyHealth@EU Consumer	Display Name	Coding Scheme	Node Representation
Patient Summary	Patient Summary	2.16.840.1.113883.6.1	60591-5
eDispensation	eDispensation	2.16.840.1.113883.6.1	60593-1
ePrescription	ePrescription	2.16.840.1.113883.6.1	57833-6
eDispensation Discard	eDispensation Discard	2.16.840.1.113883.6.1	DISCARD-60593-1

6.3 WSDLs

This section lists the WSDLs for the MyHealth@EU messages. The schemas for the messages and contained data types are available at the [eHDSI Technical Guidelines].

6.4 MyHealth@EU Error Codes

Errors exchanged between NCPs MUST follow an agreed standardization, independently of the NCPs being OpenNCP or a national implementation".

The standardized exceptions can be divided into 2 categories:

"error": when the exception prevents the proper continuity of the exchanges between the Country of Affiliation and the Country of Treatment

"warning": when the exception does not prevent the proper continuity of the exchanges between the Country of Affiliation and the Country of Treatment

The exception codes are constructed by following the format:
[ERROR_LEVEL]_[SYNTHETIC_DESCRIPTION_OF_THE_ERROR]

List of the errors exchanged between NCP are described on the Confluence page on Exception Handling in MyHealth@EU:
https://webgate.ec.europa.eu/fpfis/wikis/x/HIfdPw

The following error codes are defined for MyHealth@EU services and MUST be supported by MyHealth@EU for backward compatibility. New implementations of MyHealth@EU NCP-A MUST NOT issue these errors.

Warning Condition and Severity	Response Status	MyHealth@EU Warning Message (codeContext attribute)	Code (errorCode attribute)	Target
If no format qualifier is given: Not all of the requested encodings are provided (e.g. due to inability to transcode a certain national code). (ERROR)	PartialSuccess	Rendering incomplete	4101	ID of the provided ePrescription document that is missing alternative encodings
If MyHealth@EU pivot CDA format is requested: NCP-A cannot provide the minimum dataset for all its registered ePrescriptions. The HP MAY request the source coded PDF. (ERROR)	PartialSuccess	Collection incomplete	4102	None
The HP MUST consider additionally the source coded document because it MAY contain information that is not included in the MyHealth@EU pivot CDA (e.g. because field were nullified due to missing code mappings) (WARNING)	Success	Source coded document must be considered	2102	IDs of the affected documents
The prescribed medication has not been checked for interdependencies with the patient's current medication (e.g. because of country A legal restrictions). (WARNING)	Success	Dependencies not checked	2104	None
The prescription is available for dispensation but not valid for reimbursement. (WARNING)	Success	No reimbursement	2105	IDs of the affected ePrescriptions
eDisensations were received but not processed	Success	Processing deferred	2201	None

Table 1 : Error Code

Condition and Severity	Response Status	Message	Code	Action to be taken
The patient has not given consent to the requested service. (ERROR)	Failure	No Consent	4701	The HP SHOULD ask the patient to give consent to the requested service in country B. If the patient gives consent, the consent MUST be transmitted to country-A by using the

				respective operation of the MyHealth@EU consent service. If such consent giving procedure is accepted by country A, HP SHOULD re-issue the request for medical data.
Country A requests a higher authentication trust level than assigned to the HP (e.g. password-based login is not accepted for the requested operation). (ERROR)	Failure	Weak Authentication	4702	If possible, the HP SHOULD log in again with a stronger mechanism (e.g. smartcard) and re-issue the request with the respective identity assertion.
Either the security policy of country A or a privacy policy of the patient (that was given in country A) does not allow the requested operation to be performed by the HP (ERROR).	Failure	Insufficient Rights	4703	If the HP can switch to another (appropriate) role, he SHOULD do so and re-issue the request.
No document of the requested kind is registered for the given patient. (WARNING)	Success	No Data	1101 (eP) or 1102 (PS) or 1104 (OrCD)	
If PDF-coded document is requested: Country A does not provide the (optional) source coded version of the document (INFO)	Success	Unsupported Feature	4201	The service consumer SHOULD re-issue the request with another encoding.
The query argument slots used by the service consumer are not supported by the service provider. (ERROR)	Failure	Unknown Signifier	4202	The service consumer SHOULD re-issue the request with another set of query arguments.
The requested encoding cannot be provided due to a transcoding error. (ERROR)	Failure	Transcoding Error	4203	The service consumer SHOULD re-issue the request with another encoding.
The service provider is unable to evaluate the given argument values (ERROR).	Failure	Unknown Filter	4204	The service consumer MAY re-issue the request using another semantic signifier.
The service provider does not implement the On Demand Documents option as requested by the service consumer.	Failure	Unknown Option	4205	The service consumer SHOULD re-issue the request without using the On Demand Documents option.
The patient identifier in the query differs from the patient identifier in the TRC assertion (see chapter 4.2.1).	Failure	Unknown Patient Identifier	4206	
The requested encoding cannot be provided due to a transcoding error. (ERROR)	Failure	Transcoding Error	4203	The service consumer SHOULD re-issue the request with another encoding.
The extra parameters linked to the document identifier in the query differs from the ones in the TRC assertion (see chapter 4.2.2).	Failure	Unknow Document Identifier	4207	The extra parameters linked to the document identifier in the query differs from the ones in the TRC assertion (see chapter 4.2.2).

Table 2 : Error Code

Condition and Severity	Location	Message	Code	Action to be taken
No matching ePrescription was found (ERROR)	OID of the eDispensation document that caused the error.	No match	4105	HP-B (or NCP-B depending on the concrete implementation) SHOULD check the document IDs and re-issue the request.

ePrescription has already been dispensed (ERROR)	OID of the eDispensation document that caused the error.	Invalid Dispensation	4106	HP-B SHOULD again query for the list of available ePrescription.
Country A requests a higher authen- tication trust level than assigned to the HP (e.g. password-based login is not accepted for the requested operation). (ERROR)	-	Weak Authentication	4702	If possible, the HP SHOULD log in again with a stronge mechanism (e.g. smartcard) and re-issue the request with the respective identity assertion.
The eDispensation service provider only accepts dispensation data that is digitally signed by an HP. (ERROR)	OID of the eDispensation document that caused the error.	No Signature	4704	If possible, NCP-B SHOULD re-issue the request with the data signed by an HP.
The service consumer did not provide the MyHealth@EU pivot coded document for an eDispensation (ERROR)	OID of the source coded eDispensation that in not additionally provided as MyHealth@EU pivot coded document	Pivot data missing	4108	The source coded document MUST NOT be processed by the service provider. The service consumer MUST retransmit the dispensation with both encodings.

Table 3 : Error Code

7 References

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